

Stakeholder 17: External maintenance contractors

A practical catalogue for safe, evidence-aware maintenance decisions in a live building

1 mobilisation, induction and site information

MC-001 to MC-010

2 access, permits, escorts and work windows

MC-011 to MC-020

3 asset location, drawings, manuals and history

MC-021 to MC-030

4 fault context, diagnosis support and work orders

MC-031 to MC-040

5 tools, spares, deliveries and logistics

MC-041 to MC-050

6 isolations, hazards, method statements and safe boundaries

MC-051 to MC-060

7 testing, evidence, completion and handover

MC-061 to MC-070

8 disruption, coordination, waste and service restoration

MC-071 to MC-080

Scope and safe-use conditions

What this catalogue can support - and what it deliberately does not do.

Purpose

Practical questions for external maintenance contractors preparing, attending, diagnosing, maintaining, testing and handing back work in Abacws. This is a question-and-evidence design artefact, not an operating instruction, permit system or substitute for competent judgement.

Proposed evidence only

Sensors, telemetry, integrations and analytics are proposals until deployment, asset/location mapping, commissioning, permission, time alignment and current health are independently verified. Every answer exposes spatial coverage, observation time, sampling/aggregation and quality limits.

Current authority wins

Facilities/Estates, Security, Health and Safety, Fire Safety, permit issuers, Authorised Persons, service operators, engineers and document owners remain authoritative within their responsibilities. Current controlled owner decisions take precedence over analytics and sensor indications.

Historical spatial reference

The May 2021 floor plan is historical orientation only. It cannot establish current room use, asset position, access, services, hazards, closures, storage, egress routes or work permissions. Current controlled records and competent owners always take precedence.

No authorisation or control

The service must not authorise work, access, permits, escorts, isolations, hot work, shutdowns, energisation, restart or restoration, and must not operate access controls, BMS points, switches, breakers, valves, alarms or plant.

Safety, privacy and honest failure

Do not replace RAMS, risk assessment, COSHH/asbestos processes, lockout, proving dead/zero energy, lifting plans, confined-space controls or competent supervision. Avoid person identification/tracking and restricted information. Missing, stale, conflicting or unhealthy evidence returns a bounded unknown and the responsible owner.

Field guide

Every record exposes whether an answer is technically and operationally defensible.

1 Stakeholder question

A natural contractor decision question for mobilisation, access, asset verification, diagnosis, logistics, safe work, testing or restoration.

2 Sensors and telemetry

Physically plausible proposed evidence with modality/location, coverage, sampling/aggregation, calibration/verification, clocks, uptime, gaps, drift and provenance. Unjustified sensing is explicitly rejected.

3 Authoritative supporting data

Named responsibility owners and controlled records outrank sensor inference. Effective/observed/retrieved times, permissions, provenance, staleness and conflicts stay visible.

4 Analysis required

Explicit operation type: lookup, observation, calculation, estimate, forecast, comparison, diagnosis or recommendation; plus decision interval, minimum evidence, method, uncertainty and safe failure.

5 Answer boundary

Limits on inference, disclosure, promises, certification, authorisation and control. Information never becomes permission or a substitute for competent safety processes.

6 Why this is useful

The concrete contractor action enabled: dispatch, route, identify, test, bring kit, stop/escalate, coordinate, hand over or restore under the correct authority.

Evidence, permissions and source precedence

Rules for deciding whether evidence is usable for an external maintenance decision.

Precedence chain

1) live emergency/safety/permit/Authorised Person decisions; 2) current controlled engineering/operational records; 3) verified direct observations and commissioned telemetry; 4) uncontrolled copies; 5) historical reference. Lower levels never override higher authority.

Time semantics

Separate effective time for rules/documents, observed time for physical states/events and retrieved time for the answer. Current is decision-specific; show freshness and whether the evidence covers the decision interval.

Telemetry quality gate

Use proposed sensing only after deployment, mapping, commissioning, permission and health checks. Show modality/location, sampling and aggregation, calibration/verification, units, clock offset, uptime, gaps, drift, range/detection limits and source provenance.

Minimum evidence

Data existence is not enough. Each record states what must be present before a lookup, comparison, calculation, forecast or diagnosis can support a decision.

Conflict and uncertainty

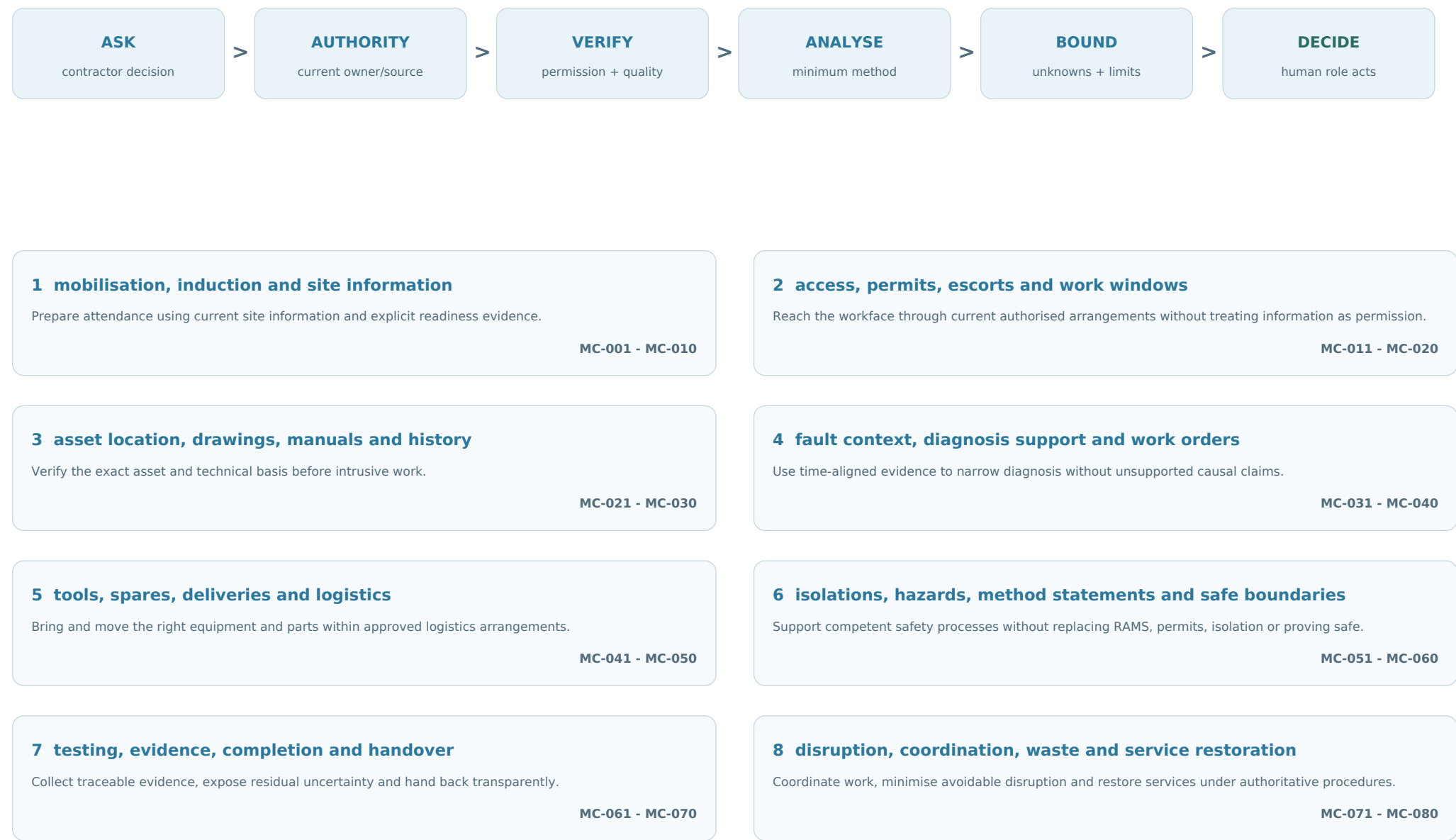
Keep conflicts visible and apply owner-defined precedence. Measurement uncertainty, incomplete coverage, confounding, stale mapping and missing intervals remain explicit rather than being hidden in a confidence score.

Safe failure

When evidence is missing, stale, conflicting, outside permission or technically unhealthy, answer "not established", name the blocking gap, state whether it blocks the decision and refer to the responsible current owner.

Catalogue map and use pattern

Ask -> authority -> verify evidence -> analyse -> expose boundary -> human role decides within its authority.



MC-001

Category 1

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Is the visit ready to mobilise, and what must our crew resolve or bring before travelling?

Sensors and telemetry

Proposed evidence: No physical sensor is required for the authoritative decision. If contextual evidence is useful, use only non-identifying door/controller health, aggregate area counts or relevant plant status. Coverage: the explicitly approved entrance, route, work zone or asset only. Timing: event time and 5-minute area summaries for the decision window. Quality gate: No sensing is needed to establish the authoritative answer. If contextual telemetry is shown, first verify deployment, asset/location mapping, commissioning, permission, source identity, clock alignment and current health; expose uptime/gaps and provenance. Profile-specific limit: verify mapping/commissioning; any count is approximate and any physical sensor used for a threshold needs current calibration/verification. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Work-order or contract owner, Facilities duty manager and contractor supervisor: current work order, mobilisation prerequisites, site instructions and attendance constraints. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + comparison. Decision interval: before travel/start and again when crew, scope, site instruction or work window changes. Minimum evidence: a current controlled source for visit readiness and mobilisation prerequisites matching the exact job, asset/location and decision time; two like-for-like records/states with common units, known configuration and comparable operating context. Method: resolve exact controlled fields/status and retain source/version; compare like-for-like quantity, units, configuration and operating context, applied to visit readiness and mobilisation prerequisites. Uncertainty: state source/version conflicts, staleness, scope, permission and any ambiguity in asset/location or applicability; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not certify induction/competence, approve RAMS or promise current site availability. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Dispatch the crew only when required prerequisites are resolved.

MC-002

Category 1

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Which planned crew members still need a valid site induction, refresher or task-specific briefing?

Sensors and telemetry

Proposed evidence: No sensing is justified for this decision. If telemetry is displayed as context, it must be limited to the exact asset/work zone and must not be used to infer people, authority, competence, contractual status or permission. Coverage: the exact asset/work zone only, and only when context is genuinely useful. Timing: current state or the shortest decision-relevant interval; no unnecessary historical collection. Quality gate: No sensing is needed to establish the authoritative answer. If contextual telemetry is shown, first verify deployment, asset/location mapping, commissioning, permission, source identity, clock alignment and current health; expose uptime/gaps and provenance. Profile-specific limit: verify mapping, commissioning, permission, clock, health and provenance; no sensor output may substitute for the authoritative record. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Health and Safety/Facilities induction owner and Security identity/access owner: official induction/competence register, role/site scope and validity dates. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup. Decision interval: before travel/start and again when crew, scope, site instruction or work window changes. Minimum evidence: a current controlled source for crew induction/briefing validity matching the exact job, asset/location and decision time. Method: resolve exact controlled fields/status and retain source/version, applied to crew induction/briefing validity. Uncertainty: state source/version conflicts, staleness, scope, permission and any ambiguity in asset/location or applicability; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not certify induction/competence, approve RAMS or promise current site availability. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Assign the correct crew and arrange any missing briefing before attendance.

MC-003

Category 1

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Who owns this job on site, who is the technical contact, and who is the fallback escalation?

Sensors and telemetry

Proposed evidence: No sensing is justified for this decision. If telemetry is displayed as context, it must be limited to the exact asset/work zone and must not be used to infer people, authority, competence, contractual status or permission. Coverage: the exact asset/work zone only, and only when context is genuinely useful. Timing: current state or the shortest decision-relevant interval; no unnecessary historical collection. Quality gate: No sensing is needed to establish the authoritative answer. If contextual telemetry is shown, first verify deployment, asset/location mapping, commissioning, permission, source identity, clock alignment and current health; expose uptime/gaps and provenance. Profile-specific limit: verify mapping, commissioning, permission, clock, health and provenance; no sensor output may substitute for the authoritative record. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Work-order owner, contract administrator and Facilities duty manager: current work order, approved role directory and duty/escalation rota. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup. Decision interval: before travel/start and again when crew, scope, site instruction or work window changes. Minimum evidence: a current controlled source for on-site job ownership and escalation matching the exact job, asset/location and decision time. Method: resolve exact controlled fields/status and retain source/version, applied to on-site job ownership and escalation. Uncertainty: state source/version conflicts, staleness, scope, permission and any ambiguity in asset/location or applicability; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not certify induction/competence, approve RAMS or promise current site availability. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Know the correct technical/duty contact and escalation route without relying on guessed availability.

MC-004

Category 1

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

What emergency arrangements and temporary impairments apply to our exact work area and shift?

Sensors and telemetry

Proposed evidence: No sensing is justified for this decision. If telemetry is displayed as context, it must be limited to the exact asset/work zone and must not be used to infer people, authority, competence, contractual status or permission. Coverage: the exact asset/work zone only, and only when context is genuinely useful. Timing: current state or the shortest decision-relevant interval; no unnecessary historical collection. Quality gate: No sensing is needed to establish the authoritative answer. If contextual telemetry is shown, first verify deployment, asset/location mapping, commissioning, permission, source identity, clock alignment and current health; expose uptime/gaps and provenance. Profile-specific limit: verify mapping, commissioning, permission, clock, health and provenance; no sensor output may substitute for the authoritative record. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Fire Safety, Health and Safety, Security and building duty manager: current emergency plan, temporary impairment notices and contractor briefing. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup. Decision interval: before travel/start and again when crew, scope, site instruction or work window changes. Minimum evidence: a current controlled source for area/time-specific emergency arrangements matching the exact job, asset/location and decision time. Method: resolve exact controlled fields/status and retain source/version, applied to area/time-specific emergency arrangements. Uncertainty: state source/version conflicts, staleness, scope, permission and any ambiguity in asset/location or applicability; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not certify induction/competence, approve RAMS or promise current site availability. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Brief the crew on current emergency arrangements and raise any impairment before starting.

MC-005

Category 1

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Which current site rules change our RAMS, tooling, sequence, containment or housekeeping for this task?

Sensors and telemetry

Proposed evidence: No physical sensor is required for the authoritative decision. If contextual evidence is useful, use only non-identifying door/controller health, aggregate area counts or relevant plant status. Coverage: the explicitly approved entrance, route, work zone or asset only. Timing: event time and 5-minute area summaries for the decision window. Quality gate: No sensing is needed to establish the authoritative answer. If contextual telemetry is shown, first verify deployment, asset/location mapping, commissioning, permission, source identity, clock alignment and current health; expose uptime/gaps and provenance. Profile-specific limit: verify mapping/commissioning; any count is approximate and any physical sensor used for a threshold needs current calibration/verification. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Facilities site-rules owner, Health and Safety and local service/area owner: current contractor rules, task restrictions and location-specific controls. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + recommendation. Decision interval: before travel/start and again when crew, scope, site instruction or work window changes. Minimum evidence: a current controlled source for task-applicable site rules matching the exact job, asset/location and decision time; only options already permitted by the responsible owner, with explicit ranking criteria. Method: resolve exact controlled fields/status and retain source/version; rank only owner-permitted options by explicit criteria; recommendation creates no permission, applied to task-applicable site rules. Uncertainty: state source/version conflicts, staleness, scope, permission and any ambiguity in asset/location or applicability; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not certify induction/competence, approve RAMS or promise current site availability. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Adapt tooling, sequencing and housekeeping to current site rules before reaching the workforce.

MC-006

Category 1

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Which contractor welfare and break facilities are available during our shift, and where are they?

Sensors and telemetry

Proposed evidence: No sensing is justified for this decision. If telemetry is displayed as context, it must be limited to the exact asset/work zone and must not be used to infer people, authority, competence, contractual status or permission. Coverage: the exact asset/work zone only, and only when context is genuinely useful. Timing: current state or the shortest decision-relevant interval; no unnecessary historical collection. Quality gate: No sensing is needed to establish the authoritative answer. If contextual telemetry is shown, first verify deployment, asset/location mapping, commissioning, permission, source identity, clock alignment and current health; expose uptime/gaps and provenance. Profile-specific limit: verify mapping, commissioning, permission, clock, health and provenance; no sensor output may substitute for the authoritative record. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Facilities/site-services owner and local duty manager: current contractor welfare designation, opening status and temporary closures. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + observation. Decision interval: before travel/start and again when crew, scope, site instruction or work window changes. Minimum evidence: a current controlled source for designated contractor welfare availability matching the exact job, asset/location and decision time; a direct observation or healthy mapped measurement covering the required point/window. Method: resolve exact controlled fields/status and retain source/version; time-align only relevant observed/measured states and report observation, not cause, applied to designated contractor welfare availability. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not certify induction/competence, approve RAMS or promise current site availability. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Plan welfare and breaks using confirmed current facilities rather than historical assumptions.

MC-007

Category 1

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Has any current building, plant or access condition invalidated an assumption in the work order or RAMS?

Sensors and telemetry

Proposed evidence: Proposed mapped plant alarms plus temperature, pressure, flow, current, position or equipment-state telemetry can corroborate current process condition. Coverage: target asset and directly connected upstream/downstream points only. Timing: native alarm events plus 1-minute process data, with 5-minute display summaries where useful. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify point-to-asset mapping, units, calibration/verification, clock offset, uptime, gaps, dead bands, saturation and drift. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Work-order owner, local service operator and Facilities duty manager: current plant/access condition, active incidents, impairments and work-order assumptions. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: comparison + observation. Decision interval: before travel/start and again when crew, scope, site instruction or work window changes. Minimum evidence: a current controlled source for changes from work-order assumptions matching the exact job, asset/location and decision time; a direct observation or healthy mapped measurement covering the required point/window; two like-for-like records/states with common units, known configuration and comparable operating context. Method: compare like-for-like quantity, units, configuration and operating context; time-align only relevant observed/measured states and report observation, not cause, applied to changes from work-order assumptions. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not certify induction/competence, approve RAMS or promise current site availability. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Stop and re-check the method when live conditions no longer match planning assumptions.

MC-008

Category 1

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Which job information is controlled and current, and which is historical, uncontrolled or still unverified?

Sensors and telemetry

Proposed evidence: No physical sensor can establish document authority. Telemetry metadata may only corroborate which mapped asset/location produced a reading cited beside a controlled record. Coverage: only the asset/location referenced by the current job document. Timing: source metadata at retrieval time; no physical sampling is required. Quality gate: No sensing is needed to establish the authoritative answer. If contextual telemetry is shown, first verify deployment, asset/location mapping, commissioning, permission, source identity, clock alignment and current health; expose uptime/gaps and provenance. Profile-specific limit: verify point/source ID, mapping version and pipeline health if telemetry is cited. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Document controller, asset-data owner and work-order owner: controlled drawings/manuals, version/status, supersession information and approved repositories. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + comparison. Decision interval: before travel/start and again when crew, scope, site instruction or work window changes. Minimum evidence: a current controlled source for document authority and currency matching the exact job, asset/location and decision time; two like-for-like records/states with common units, known configuration and comparable operating context. Method: resolve exact controlled fields/status and retain source/version; compare like-for-like quantity, units, configuration and operating context, applied to document authority and currency. Uncertainty: state source/version conflicts, staleness, scope, permission and any ambiguity in asset/location or applicability; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not certify induction/competence, approve RAMS or promise current site availability. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Take only controlled/current information to the job and identify what still needs owner confirmation.

MC-009

Category 1

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Which specialist competence, formal appointment or named technical role must be present for this scope?

Sensors and telemetry

Proposed evidence: No sensing is justified for this decision. If telemetry is displayed as context, it must be limited to the exact asset/work zone and must not be used to infer people, authority, competence, contractual status or permission. Coverage: the exact asset/work zone only, and only when context is genuinely useful. Timing: current state or the shortest decision-relevant interval; no unnecessary historical collection. Quality gate: No sensing is needed to establish the authoritative answer. If contextual telemetry is shown, first verify deployment, asset/location mapping, commissioning, permission, source identity, clock alignment and current health; expose uptime/gaps and provenance. Profile-specific limit: verify mapping, commissioning, permission, clock, health and provenance; no sensor output may substitute for the authoritative record. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Health and Safety and the Authorised Person/competence-framework owner: current role requirements, formal appointments and task/permit scope. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup. Decision interval: before travel/start and again when crew, scope, site instruction or work window changes. Minimum evidence: a current controlled source for required specialist roles/appointments matching the exact job, asset/location and decision time. Method: resolve exact controlled fields/status and retain source/version, applied to required specialist roles/appointments. Uncertainty: state source/version conflicts, staleness, scope, permission and any ambiguity in asset/location or applicability; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not certify induction/competence, approve RAMS or promise current site availability. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Ensure the required competent roles are planned before a controlled task begins.

MC-010

Category 1

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

What extra staffing, access, supervision or service constraints apply to this out-of-hours visit?

Sensors and telemetry

Proposed evidence: Proposed door/hatch contacts, access-controller health, lift status and privacy-preserving area counts can corroborate route condition; they cannot grant access or infer entitlement. Coverage: approved entrance/route segments and the destination work zone only. Timing: door/lift events in real time and area counts as 5-minute aggregates. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify point mapping, controller health, clock alignment, count accuracy and sensor uptime. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Facilities duty manager, Security, service operator and permit owner as applicable: current out-of-hours staffing, access, supervision, escalation and shutdown constraints. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + comparison. Decision interval: before travel/start and again when crew, scope, site instruction or work window changes. Minimum evidence: a current controlled source for out-of-hours controls and support matching the exact job, asset/location and decision time; two like-for-like records/states with common units, known configuration and comparable operating context. Method: resolve exact controlled fields/status and retain source/version; compare like-for-like quantity, units, configuration and operating context, applied to out-of-hours controls and support. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not certify induction/competence, approve RAMS or promise current site availability. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Plan staffing, supervision and support so out-of-hours work is not stranded.

MC-011

Category 2

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Which authorised entrance and route should we use for our crew and the actual equipment we are bringing?

Sensors and telemetry

Proposed evidence: Proposed door/hatch contacts, access-controller health, lift status and privacy-preserving area counts can corroborate route condition; they cannot grant access or infer entitlement. Coverage: approved entrance/route segments and the destination work zone only. Timing: door/lift events in real time and area counts as 5-minute aggregates. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify point mapping, controller health, clock alignment, count accuracy and sensor uptime. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Security and Facilities/site-logistics owner: approved entrance, route, lift/door constraints and temporary closures for the stated load/equipment. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + recommendation. Decision interval: before access/movement and again after any permit, route, escort or window change. Minimum evidence: a current controlled source for approved entrance and equipment route matching the exact job, asset/location and decision time; only options already permitted by the responsible owner, with explicit ranking criteria. Method: resolve exact controlled fields/status and retain source/version; rank only owner-permitted options by explicit criteria; recommendation creates no permission, applied to approved entrance and equipment route. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not grant access, issue/extend permits, waive escort/route controls or create a work window from low occupancy. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Use the correct entrance and route for the actual equipment being moved.

MC-012

Category 2

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Which badge, key, token or other credential is required at each controlled point, and how are access failures escalated?

Sensors and telemetry

Proposed evidence: No sensing is justified for this decision. If telemetry is displayed as context, it must be limited to the exact asset/work zone and must not be used to infer people, authority, competence, contractual status or permission. Coverage: the exact asset/work zone only, and only when context is genuinely useful. Timing: current state or the shortest decision-relevant interval; no unnecessary historical collection. Quality gate: No sensing is needed to establish the authoritative answer. If contextual telemetry is shown, first verify deployment, asset/location mapping, commissioning, permission, source identity, clock alignment and current health; expose uptime/gaps and provenance. Profile-specific limit: verify mapping, commissioning, permission, clock, health and provenance; no sensor output may substitute for the authoritative record. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Security access-control owner: current credential type, validity/scope, controlled points and access-failure escalation procedure. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup. Decision interval: before access/movement and again after any permit, route, escort or window change. Minimum evidence: a current controlled source for credential requirement and access escalation matching the exact job, asset/location and decision time. Method: resolve exact controlled fields/status and retain source/version, applied to credential requirement and access escalation. Uncertainty: state source/version conflicts, staleness, scope, permission and any ambiguity in asset/location or applicability; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not grant access, issue/extend permits, waive escort/route controls or create a work window from low occupancy. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Bring the right credential and resolve failures through Security rather than insecure workarounds.

MC-013

Category 2

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Which permit-to-work processes may apply, and which competent role can issue, suspend or cancel each permit?**Sensors and telemetry**

Proposed evidence: No telemetry can show that a permit has been issued. Gas, temperature, pressure, fire-system or electrical measurements are evidence only when the controlled permit/RAMS explicitly requires them. Coverage: the exact workforce and system boundary defined by the permit process. Timing: sampling only at the cadence specified by the controlled process. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: use traceable instruments where specified; building telemetry remains contextual unless the competent permit owner accepts it. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Permit-system owner, Health and Safety and relevant Authorised Person: current permit rules, live permit register and issuer/canceller appointments. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup. Decision interval: before access/movement and again after any permit, route, escort or window change. Minimum evidence: a current controlled source for applicable permit process and issuing role matching the exact job, asset/location and decision time. Method: resolve exact controlled fields/status and retain source/version, applied to applicable permit process and issuing role. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not grant access, issue/extend permits, waive escort/route controls or create a work window from low occupancy. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Identify the permit pathway and competent issuer before the work window is lost.

MC-014

Category 2

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Where is an escort required, for how long, and how is escort cover arranged for our work window?**Sensors and telemetry**

Proposed evidence: No physical sensor is required for the authoritative decision. If contextual evidence is useful, use only non-identifying door/controller health, aggregate area counts or relevant plant status. Coverage: the explicitly approved entrance, route, work zone or asset only. Timing: event time and 5-minute area summaries for the decision window. Quality gate: No sensing is needed to establish the authoritative answer. If contextual telemetry is shown, first verify deployment, asset/location mapping, commissioning, permission, source identity, clock alignment and current health; expose uptime/gaps and provenance. Profile-specific limit: verify mapping/commissioning; any count is approximate and any physical sensor used for a threshold needs current calibration/verification. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Security or controlled-area owner and work-order owner: current escort requirement, escort booking/availability and permitted duration/scope. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup. Decision interval: before access/movement and again after any permit, route, escort or window change. Minimum evidence: a current controlled source for escort requirement and arrangement matching the exact job, asset/location and decision time. Method: resolve exact controlled fields/status and retain source/version, applied to escort requirement and arrangement. Uncertainty: state source/version conflicts, staleness, scope, permission and any ambiguity in asset/location or applicability; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not grant access, issue/extend permits, waive escort/route controls or create a work window from low occupancy. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Book required escort support and sequence work around realistic escorted access.

MC-015

Category 2

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

What is the approved work window, and which teaching, event, operational or blackout periods must we avoid?

Sensors and telemetry

Proposed evidence: No sensing is justified for this decision. If telemetry is displayed as context, it must be limited to the exact asset/work zone and must not be used to infer people, authority, competence, contractual status or permission. Coverage: the exact asset/work zone only, and only when context is genuinely useful. Timing: current state or the shortest decision-relevant interval; no unnecessary historical collection. Quality gate: No sensing is needed to establish the authoritative answer. If contextual telemetry is shown, first verify deployment, asset/location mapping, commissioning, permission, source identity, clock alignment and current health; expose uptime/gaps and provenance. Profile-specific limit: verify mapping, commissioning, permission, clock, health and provenance; no sensor output may substitute for the authoritative record. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Work-order owner, local area/teaching-event scheduler and Facilities operations: approved work window, blackouts and service restrictions. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + recommendation. Decision interval: before access/movement and again after any permit, route, escort or window change. Minimum evidence: a current controlled source for authorised work window and blackout periods matching the exact job, asset/location and decision time; only options already permitted by the responsible owner, with explicit ranking criteria. Method: resolve exact controlled fields/status and retain source/version; rank only owner-permitted options by explicit criteria; recommendation creates no permission, applied to authorised work window and blackout periods. Uncertainty: state source/version conflicts, staleness, scope, permission and any ambiguity in asset/location or applicability; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not grant access, issue/extend permits, waive escort/route controls or create a work window from low occupancy. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Schedule disruptive activity only inside owner-approved windows and blackout constraints.

MC-016

Category 2

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

What current controls apply to reaching or working in the ceiling void, riser, roof, plant room or floor void we need?

Sensors and telemetry

Proposed evidence: Proposed door/hatch contacts, access-controller health, lift status and privacy-preserving area counts can corroborate route condition; they cannot grant access or infer entitlement. Coverage: approved entrance/route segments and the destination work zone only. Timing: door/lift events in real time and area counts as 5-minute aggregates. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify point mapping, controller health, clock alignment, count accuracy and sensor uptime. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Facilities asset/area owner, Health and Safety and permit owner: current special-access requirements for roof, void, riser, plant room or floor void. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + observation. Decision interval: before access/movement and again after any permit, route, escort or window change. Minimum evidence: a current controlled source for restricted-space/roof/void access controls matching the exact job, asset/location and decision time; a direct observation or healthy mapped measurement covering the required point/window. Method: resolve exact controlled fields/status and retain source/version; time-align only relevant observed/measured states and report observation, not cause, applied to restricted-space/roof/void access controls. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not grant access, issue/extend permits, waive escort/route controls or create a work window from low occupancy. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Arrive with the controls and competent support needed for roof/void/plant-room access.

MC-017

Category 2

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

What restrictions apply today to noisy, dusty, odorous, vibration-producing or hot work in this area?

Sensors and telemetry

Proposed evidence: Proposed temperature, humidity, sound level, PM2.5/PM10 and, only where chemically meaningful, VOC sensing can describe local physical conditions without identifying people. Coverage: workface plus owner-defined adjacent boundary/receptor points only. Timing: 1-minute samples displayed as 5-minute aggregates; preserve exceedance events. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: threshold decisions require task-appropriate calibrated instruments; state range, detection limit, humidity/cross-sensitivity, drift and placement limits. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Local area owner, Health and Safety, Fire Safety and work-coordination owner as applicable: current noisy/dusty/odorous/vibration/hot-work restrictions and containment requirements. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + observation + comparison. Decision interval: before access/movement and again after any permit, route, escort or window change. Minimum evidence: a current controlled source for noise/dust/odour/hot-work restrictions matching the exact job, asset/location and decision time; a direct observation or healthy mapped measurement covering the required point/window; two like-for-like records/states with common units, known configuration and comparable operating context. Method: resolve exact controlled fields/status and retain source/version; time-align only relevant observed/measured states and report observation, not cause; compare like-for-like quantity, units, configuration and operating context, applied to noise/dust/odour/hot-work restrictions. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not grant access, issue/extend permits, waive escort/route controls or create a work window from low occupancy. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Choose containment, timing and monitoring appropriate to the local restriction.

MC-018

Category 2

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

If the work overruns, when must we stop, whom must we contact, and what must be re-authorised before continuing?

Sensors and telemetry

Proposed evidence: No physical sensor is required for the authoritative decision. If contextual evidence is useful, use only non-identifying door/controller health, aggregate area counts or relevant plant status. Coverage: the explicitly approved entrance, route, work zone or asset only. Timing: event time and 5-minute area summaries for the decision window. Quality gate: No sensing is needed to establish the authoritative answer. If contextual telemetry is shown, first verify deployment, asset/location mapping, commissioning, permission, source identity, clock alignment and current health; expose uptime/gaps and provenance. Profile-specific limit: verify mapping/commissioning; any count is approximate and any physical sensor used for a threshold needs current calibration/verification. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Work-order owner, duty manager, permit issuer and Security as applicable: continuation rules, expiry times and escalation contacts. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + recommendation. Decision interval: before access/movement and again after any permit, route, escort or window change. Minimum evidence: a current controlled source for overrun escalation before window expiry matching the exact job, asset/location and decision time; only options already permitted by the responsible owner, with explicit ranking criteria. Method: resolve exact controlled fields/status and retain source/version; rank only owner-permitted options by explicit criteria; recommendation creates no permission, applied to overrun escalation before window expiry. Uncertainty: state source/version conflicts, staleness, scope, permission and any ambiguity in asset/location or applicability; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not grant access, issue/extend permits, waive escort/route controls or create a work window from low occupancy. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Escalate an overrun early enough to obtain an authorised continuation or controlled pause.

MC-019

Category 2

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Can the approved route safely accommodate our ladders, trolley, test gear and replacement equipment?

Sensors and telemetry

Proposed evidence: Proposed door/lift state, lift load/fault status and temporary laser/ultrasonic clearance measurements can corroborate equipment-route constraints. Coverage: every limiting segment of the already approved route. Timing: state at movement time and one-off clearance measurements at actual constraints. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify route mapping, lift status and clearance-tool accuracy/calibration; state tolerance and temporary-obstruction uncertainty. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Facilities/site-logistics owner and lifting/manual-handling competent role where applicable: approved route, measured clearances, floor/lift capacities and handling constraints. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: calculation + comparison. Decision interval: before access/movement and again after any permit, route, escort or window change. Minimum evidence: a current controlled source for equipment-route clearance/capacity matching the exact job, asset/location and decision time; two like-for-like records/states with common units, known configuration and comparable operating context; all calculation inputs, units, capacities/tolerances and source versions. Method: show arithmetic, units and tolerance/capacity basis; compare like-for-like quantity, units, configuration and operating context, applied to equipment-route clearance/capacity. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not grant access, issue/extend permits, waive escort/route controls or create a work window from low occupancy. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Choose handling, route or lifting support that fits verified clearance/capacity constraints.

MC-020

Category 2

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Are any entrances, lifts, corridors or work zones on our route temporarily closed, restricted or shared with other work?

Sensors and telemetry

Proposed evidence: Proposed door/hatch contacts, access-controller health, lift status and privacy-preserving area counts can corroborate route condition; they cannot grant access or infer entitlement. Coverage: approved entrance/route segments and the destination work zone only. Timing: door/lift events in real time and area counts as 5-minute aggregates. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify point mapping, controller health, clock alignment, count accuracy and sensor uptime. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Security, Facilities operations and work-coordination owner: live closures, route restrictions, shared work areas and temporary barriers. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + observation. Decision interval: before access/movement and again after any permit, route, escort or window change. Minimum evidence: a current controlled source for temporary route closures and shared-work controls matching the exact job, asset/location and decision time; a direct observation or healthy mapped measurement covering the required point/window. Method: resolve exact controlled fields/status and retain source/version; time-align only relevant observed/measured states and report observation, not cause, applied to temporary route closures and shared-work controls. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not grant access, issue/extend permits, waive escort/route controls or create a work window from low occupancy. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Avoid blocked routes and conflicts with another controlled work area.

MC-021

Category 3

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Where is the exact maintainable asset, and what checks confirm we have the correct item before touching it?

Sensors and telemetry

Proposed evidence: A deliberate QR/NFC/RFID asset-tag read, nameplate check or controller identity can corroborate the exact equipment variant; continuous tracking is not required. Coverage: target asset and its immediate maintainable assembly only. Timing: one-off field read at the workface plus current controller identity where relevant. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify tag-to-asset mapping, identifier legibility, controller/source version and provenance. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Asset-register owner, document controller and work-order owner: current asset ID/location, parent-child relation, make/model and controlled job reference. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + observation + comparison. Decision interval: before using the technical basis or touching the asset, and after any field discrepancy. Minimum evidence: a current controlled source for target asset identity matching the exact job, asset/location and decision time; a direct observation or healthy mapped measurement covering the required point/window; two like-for-like records/states with common units, known configuration and comparable operating context. Method: resolve exact controlled fields/status and retain source/version; time-align only relevant observed/measured states and report observation, not cause; compare like-for-like quantity, units, configuration and operating context, applied to target asset identity. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not treat historical plans, approximate location or similar equipment as proof of current asset/service identity. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Prevent work on the wrong asset by establishing a defensible identity at the workface.

MC-022

Category 3

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Which controlled drawing is current for this asset and the services relevant to today's task?

Sensors and telemetry

Proposed evidence: No physical sensor can establish document authority. Telemetry metadata may only corroborate which mapped asset/location produced a reading cited beside a controlled record. Coverage: only the asset/location referenced by the current job document. Timing: source metadata at retrieval time; no physical sampling is required. Quality gate: No sensing is needed to establish the authoritative answer. If contextual telemetry is shown, first verify deployment, asset/location mapping, commissioning, permission, source identity, clock alignment and current health; expose uptime/gaps and provenance. Profile-specific limit: verify point/source ID, mapping version and pipeline health if telemetry is cited. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Document controller and engineering/service owner: controlled drawings, version/status, supersession record, redlines/as-builts and approved changes. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + comparison. Decision interval: before using the technical basis or touching the asset, and after any field discrepancy. Minimum evidence: a current controlled source for current controlled drawing matching the exact job, asset/location and decision time; two like-for-like records/states with common units, known configuration and comparable operating context. Method: resolve exact controlled fields/status and retain source/version; compare like-for-like quantity, units, configuration and operating context, applied to current controlled drawing. Uncertainty: state source/version conflicts, staleness, scope, permission and any ambiguity in asset/location or applicability; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not treat historical plans, approximate location or similar equipment as proof of current asset/service identity. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Use the correct controlled drawing and know which parts still require field verification.

MC-023

Category 3

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Which manufacturer and maintenance instructions apply to the exact installed model, variant and configuration?

Sensors and telemetry

Proposed evidence: A deliberate QR/NFC/RFID asset-tag read, nameplate check or controller identity can corroborate the exact equipment variant; continuous tracking is not required. Coverage: target asset and its immediate maintainable assembly only. Timing: one-off field read at the workface plus current controller identity where relevant. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify tag-to-asset mapping, identifier legibility, controller/source version and provenance. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Asset owner, OEM/manual repository owner and document controller: current approved operation and maintenance instructions, manufacturer bulletins and exact installed variant. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + comparison. Decision interval: before using the technical basis or touching the asset, and after any field discrepancy. Minimum evidence: a current controlled source for exact installed-variant instructions matching the exact job, asset/location and decision time; two like-for-like records/states with common units, known configuration and comparable operating context. Method: resolve exact controlled fields/status and retain source/version; compare like-for-like quantity, units, configuration and operating context, applied to exact installed-variant instructions. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not treat historical plans, approximate location or similar equipment as proof of current asset/service identity. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Bring the exact instructions for the installed variant rather than a similar model.

MC-024

Category 3

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Which maintenance, failure and remedial history is relevant to this job, without assuming past events caused today's fault?

Sensors and telemetry

Proposed evidence: No physical sensor can establish document authority. Telemetry metadata may only corroborate which mapped asset/location produced a reading cited beside a controlled record. Coverage: only the asset/location referenced by the current job document. Timing: source metadata at retrieval time; no physical sampling is required. Quality gate: No sensing is needed to establish the authoritative answer. If contextual telemetry is shown, first verify deployment, asset/location mapping, commissioning, permission, source identity, clock alignment and current health; expose uptime/gaps and provenance. Profile-specific limit: verify point/source ID, mapping version and pipeline health if telemetry is cited. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Asset engineer, work-order owner and maintenance-record owner: controlled maintenance history, fault records, prior interventions and verified outcomes. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + comparison. Decision interval: before using the technical basis or touching the asset, and after any field discrepancy. Minimum evidence: a current controlled source for relevant maintenance/failure history matching the exact job, asset/location and decision time; two like-for-like records/states with common units, known configuration and comparable operating context. Method: resolve exact controlled fields/status and retain source/version; compare like-for-like quantity, units, configuration and operating context, applied to relevant maintenance/failure history. Uncertainty: state source/version conflicts, staleness, scope, permission and any ambiguity in asset/location or applicability; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not treat historical plans, approximate location or similar equipment as proof of current asset/service identity. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Target relevant known history without assuming past action caused the current outcome.

MC-025

Category 3

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

What are the verified asset ID, manufacturer, model, serial number, duty and critical configuration details?

Sensors and telemetry

Proposed evidence: A deliberate QR/NFC/RFID asset-tag read, nameplate check or controller identity can corroborate the exact equipment variant; continuous tracking is not required. Coverage: target asset and its immediate maintainable assembly only. Timing: one-off field read at the workface plus current controller identity where relevant. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify tag-to-asset mapping, identifier legibility, controller/source version and provenance. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Asset-register owner and asset engineer: current identifier, manufacturer/model/serial, duty, parent-child relation and controlled configuration. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + observation + comparison. Decision interval: before using the technical basis or touching the asset, and after any field discrepancy. Minimum evidence: a current controlled source for asset identifiers and configuration matching the exact job, asset/location and decision time; a direct observation or healthy mapped measurement covering the required point/window; two like-for-like records/states with common units, known configuration and comparable operating context. Method: resolve exact controlled fields/status and retain source/version; time-align only relevant observed/measured states and report observation, not cause; compare like-for-like quantity, units, configuration and operating context, applied to asset identifiers and configuration. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not treat historical plans, approximate location or similar equipment as proof of current asset/service identity. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Confirm the exact installed equipment before choosing parts, firmware or test criteria.

MC-026

Category 3

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

What current evidence exists for hidden services at the exact place we may drill, cut, core or open up?

Sensors and telemetry

Proposed evidence: No building IoT sensor can clear hidden services. Proposed CAT/Genny, electromagnetic tracing, GPR or thermal imaging can provide local survey evidence when selected and operated by competent people. Coverage: the exact penetration/open-up zone plus the owner-defined safety margin. Timing: task-specific scan immediately before intrusive work; preserve marked/raw results and scan time. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: use maintained/calibrated locating instruments as required; state depth/material limitations, blind zones and inaccessible areas. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Engineering/service owners, document controller and permit/Health and Safety owner: current hidden-service records, surveys, penetration/open-up controls and known hazards. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + observation + comparison. Decision interval: before using the technical basis or touching the asset, and after any field discrepancy. Minimum evidence: a current controlled source for hidden-service evidence at the penetration zone matching the exact job, asset/location and decision time; a direct observation or healthy mapped measurement covering the required point/window; two like-for-like records/states with common units, known configuration and comparable operating context. Method: resolve exact controlled fields/status and retain source/version; time-align only relevant observed/measured states and report observation, not cause; compare like-for-like quantity, units, configuration and operating context, applied to hidden-service evidence at the penetration zone. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not treat historical plans, approximate location or similar equipment as proof of current asset/service identity. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Choose a safe work position or escalate before penetrating a surface with uncertain services.

MC-027

Category 3

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Which breaker, isolator, valve, damper or other device is recorded as serving this asset, and what must be field-verified?

Sensors and telemetry

Proposed evidence: Mapped voltage/current, pressure, flow, temperature, valve/position or speed telemetry can corroborate system state but cannot prove dead, zero energy or correct isolation. Coverage: target system, candidate isolation devices and directly connected stored-energy points. Timing: native events and 1-second to 1-minute values through the competent isolation verification sequence. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify mapping, units, calibration/verification, clock, dead bands, shared feeds and loss-of-signal behaviour. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Relevant Authorised Person/service owner and document controller: current schematic/single-line/piping/instrumentation diagram, isolation schedule, labels and controlled service relationship. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + observation + comparison. Decision interval: before using the technical basis or touching the asset, and after any field discrepancy. Minimum evidence: a current controlled source for documented service/isolation relationship matching the exact job, asset/location and decision time; a direct observation or healthy mapped measurement covering the required point/window; two like-for-like records/states with common units, known configuration and comparable operating context. Method: resolve exact controlled fields/status and retain source/version; time-align only relevant observed/measured states and report observation, not cause; compare like-for-like quantity, units, configuration and operating context, applied to documented service/isolation relationship. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not treat historical plans, approximate location or similar equipment as proof of current asset/service identity. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Help the competent isolating person verify the documented service relationship without treating telemetry as safety proof.

MC-028

Category 3

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

What verified changes have been made to the asset, its connections or its immediate surroundings since the last controlled baseline?

Sensors and telemetry

Proposed evidence: No fixed sensor is required. Use deliberate workplace observation and, only where approved, a non-identifying photograph or measured dimension to record current physical condition. Coverage: the target asset/workface and the minimum adjacent area needed to explain the condition. Timing: one-off observation at the decision point; repeat only after a material change. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: record observer role, time, exact location/asset, measuring-tool ID where used and image/file provenance; do not capture people, screens, research content or security-sensitive details. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Asset engineer, document controller and change-control owner: approved modifications, redlines/as-builts, configuration changes and field discrepancies. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: comparison + observation. Decision interval: before using the technical basis or touching the asset, and after any field discrepancy. Minimum evidence: a current controlled source for changes around the asset since baseline matching the exact job, asset/location and decision time; a direct observation or healthy mapped measurement covering the required point/window; two like-for-like records/states with common units, known configuration and comparable operating context. Method: compare like-for-like quantity, units, configuration and operating context; time-align only relevant observed/measured states and report observation, not cause, applied to changes around the asset since baseline. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not treat historical plans, approximate location or similar equipment as proof of current asset/service identity. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Adjust access, tools and method to the current installation rather than an old memory or plan.

MC-029

Category 3

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Do warranty, manufacturer support or specialist service conditions constrain who may work on this asset or what may be changed?

Sensors and telemetry

Proposed evidence: No sensing is justified for this decision. If telemetry is displayed as context, it must be limited to the exact asset/work zone and must not be used to infer people, authority, competence, contractual status or permission. Coverage: the exact asset/work zone only, and only when context is genuinely useful. Timing: current state or the shortest decision-relevant interval; no unnecessary historical collection. Quality gate: No sensing is needed to establish the authoritative answer. If contextual telemetry is shown, first verify deployment, asset/location mapping, commissioning, permission, source identity, clock alignment and current health; expose uptime/gaps and provenance. Profile-specific limit: verify mapping, commissioning, permission, clock, health and provenance; no sensor output may substitute for the authoritative record. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Contract/procurement owner, asset owner and manufacturer-support owner: current warranty/support agreement, serial registration, authorised maintainer conditions and commercial permissions. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup. Decision interval: before using the technical basis or touching the asset, and after any field discrepancy. Minimum evidence: a current controlled source for warranty/support constraints matching the exact job, asset/location and decision time. Method: resolve exact controlled fields/status and retain source/version, applied to warranty/support constraints. Uncertainty: state source/version conflicts, staleness, scope, permission and any ambiguity in asset/location or applicability; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not treat historical plans, approximate location or similar equipment as proof of current asset/service identity. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Pause for warranty/specialist coordination when the current support terms require it.

MC-030

Category 3

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

What unresolved defects, recurring observations or deferred actions already exist for this asset or immediate location?

Sensors and telemetry

Proposed evidence: No physical sensor can establish document authority. Telemetry metadata may only corroborate which mapped asset/location produced a reading cited beside a controlled record. Coverage: only the asset/location referenced by the current job document. Timing: source metadata at retrieval time; no physical sampling is required. Quality gate: No sensing is needed to establish the authoritative answer. If contextual telemetry is shown, first verify deployment, asset/location mapping, commissioning, permission, source identity, clock alignment and current health; expose uptime/gaps and provenance. Profile-specific limit: verify point/source ID, mapping version and pipeline health if telemetry is cited. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Work-order/defect-register owner and asset engineer: current open defects, recurring observations, deferred actions and accepted temporary conditions. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + comparison. Decision interval: before using the technical basis or touching the asset, and after any field discrepancy. Minimum evidence: a current controlled source for open defects and deferred actions matching the exact job, asset/location and decision time; two like-for-like records/states with common units, known configuration and comparable operating context. Method: resolve exact controlled fields/status and retain source/version; compare like-for-like quantity, units, configuration and operating context, applied to open defects and deferred actions. Uncertainty: state source/version conflicts, staleness, scope, permission and any ambiguity in asset/location or applicability; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not treat historical plans, approximate location or similar equipment as proof of current asset/service identity. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Avoid duplicate work and carry open defects/deferred actions into scope and handover.

MC-031

Category 4

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

What exactly was observed, from which authorised report or system, and when did the fault begin or become unacceptable?

Sensors and telemetry

Proposed evidence: Time-aligned mapped asset telemetry, alarms and traceable portable measurements can compare states only when they represent the same physical quantity and operating context. Coverage: target asset and physically connected comparison points only. Timing: native events plus 1-minute values across the defined before/during/after or peer comparison windows. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify mapping/configuration equivalence, units, calibration/verification, clock, uptime/gaps, drift, placement and response time. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Service-desk/work-order owner and asset operator: original symptom report, onset/observed times, reporter role where permitted and source provenance. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + observation. Decision interval: during fault triage and after each material new observation or intervention. Minimum evidence: a current controlled source for fault observation and onset timeline matching the exact job, asset/location and decision time; a direct observation or healthy mapped measurement covering the required point/window. Method: resolve exact controlled fields/status and retain source/version; time-align only relevant observed/measured states and report observation, not cause, applied to fault observation and onset timeline. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not convert temporal association, anomaly scores or symptom similarity into a causal diagnosis or scope extension. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Start diagnosis from a source-labelled fault timeline rather than paraphrased symptoms.

MC-032

Category 4

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Which time-aligned alarms and abnormal measurements occurred immediately before, during and after the fault?

Sensors and telemetry

Proposed evidence: Proposed mapped plant alarms plus temperature, pressure, flow, current, position or equipment-state telemetry can corroborate current process condition. Coverage: target asset and directly connected upstream/downstream points only. Timing: native alarm events plus 1-minute process data, with 5-minute display summaries where useful. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify point-to-asset mapping, units, calibration/verification, clock offset, uptime, gaps, dead bands, saturation and drift. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Asset operator/BMS or controls owner and work-order owner: mapped alarm/event log, process measurements, controller clock/status and relevant operator notes. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: observation + comparison. Decision interval: during fault triage and after each material new observation or intervention. Minimum evidence: a current controlled source for time-aligned alarms and abnormal measurements matching the exact job, asset/location and decision time; a direct observation or healthy mapped measurement covering the required point/window; two like-for-like records/states with common units, known configuration and comparable operating context. Method: time-align only relevant observed/measured states and report observation, not cause; compare like-for-like quantity, units, configuration and operating context, applied to time-aligned alarms and abnormal measurements. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not convert temporal association, anomaly scores or symptom similarity into a causal diagnosis or scope extension. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Inspect the most informative event window and components first.

MC-033

Category 4

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Is the fault continuous, intermittent or conditional on load, schedule, weather, occupancy level or operating mode?

Sensors and telemetry

Proposed evidence: Time-aligned mapped asset telemetry, alarms and traceable portable measurements can compare states only when they represent the same physical quantity and operating context. Coverage: target asset and physically connected comparison points only. Timing: native events plus 1-minute values across the defined before/during/after or peer comparison windows. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify mapping/configuration equivalence, units, calibration/verification, clock, uptime/gaps, drift, placement and response time. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Asset operator, controls owner and work-order owner: fault events plus current schedules, load/mode records and relevant environmental context. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: comparison + estimate. Decision interval: during fault triage and after each material new observation or intervention. Minimum evidence: a current controlled source for fault recurrence by load/mode/context matching the exact job, asset/location and decision time; two like-for-like records/states with common units, known configuration and comparable operating context; at least three comparable observations/periods; never a single point presented as a range. Method: compare like-for-like quantity, units, configuration and operating context; summarise comparable observations with sample count, range and coverage, applied to fault recurrence by load/mode/context. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not convert temporal association, anomaly scores or symptom similarity into a causal diagnosis or scope extension. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Plan when to reproduce an intermittent fault and which operating conditions to check.

MC-034

Category 4

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Which rooms, assets or services are confirmed affected, and which remain only suspected?

Sensors and telemetry

Proposed evidence: Proposed mapped plant alarms plus temperature, pressure, flow, current, position or equipment-state telemetry can corroborate current process condition. Coverage: target asset and directly connected upstream/downstream points only. Timing: native alarm events plus 1-minute process data, with 5-minute display summaries where useful. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify point-to-asset mapping, units, calibration/verification, clock offset, uptime, gaps, dead bands, saturation and drift. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Service owner, Facilities operations and work-order owner: confirmed impact reports, service topology/dependencies and current affected-area status. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: observation + comparison. Decision interval: during fault triage and after each material new observation or intervention. Minimum evidence: a current controlled source for confirmed versus suspected impact area matching the exact job, asset/location and decision time; a direct observation or healthy mapped measurement covering the required point/window; two like-for-like records/states with common units, known configuration and comparable operating context. Method: time-align only relevant observed/measured states and report observation, not cause; compare like-for-like quantity, units, configuration and operating context, applied to confirmed versus suspected impact area. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not convert temporal association, anomaly scores or symptom similarity into a causal diagnosis or scope extension. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Focus diagnosis and communication on the evidenced impact area.

MC-035

Category 4

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Which maintenance, setpoint, firmware, control or building changes occurred shortly before the fault, and which are only hypotheses rather than established causes?

Sensors and telemetry

Proposed evidence: Time-aligned mapped asset telemetry, alarms and traceable portable measurements can compare states only when they represent the same physical quantity and operating context. Coverage: target asset and physically connected comparison points only. Timing: native events plus 1-minute values across the defined before/during/after or peer comparison windows. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify mapping/configuration equivalence, units, calibration/verification, clock, uptime/gaps, drift, placement and response time. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Change-control owner, controls/asset engineer and work-order owner: authorised maintenance, setpoint, firmware, control and building changes with effective times. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + comparison. Decision interval: during fault triage and after each material new observation or intervention. Minimum evidence: a current controlled source for changes preceding the fault matching the exact job, asset/location and decision time; two like-for-like records/states with common units, known configuration and comparable operating context. Method: resolve exact controlled fields/status and retain source/version; compare like-for-like quantity, units, configuration and operating context, applied to changes preceding the fault. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not convert temporal association, anomaly scores or symptom similarity into a causal diagnosis or scope extension. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Check recent changes as hypotheses without treating sequence as causation.

MC-036

Category 4

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

What verified normal performance should we compare against for this asset under the same operating conditions?

Sensors and telemetry

Proposed evidence: Mapped asset/process telemetry plus a traceable spot measurement can preserve comparable pre/post evidence when the same physical quantity, point and operating mode are used. Coverage: target asset and direct context/comparison points only. Timing: preserve raw/source-rate values around the test event and 1- or 5-minute summaries over the agreed stable interval. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify mapping, units, calibration/verification, clock, uptime/gaps, drift and operating-mode comparability. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Asset engineer/controls owner and operations owner: approved design/commissioning expectations plus verified pre-fault measurements under comparable operating conditions. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: comparison + estimate. Decision interval: during fault triage and after each material new observation or intervention. Minimum evidence: a current controlled source for verified normal-performance baseline matching the exact job, asset/location and decision time; two like-for-like records/states with common units, known configuration and comparable operating context; at least three comparable observations/periods; never a single point presented as a range. Method: compare like-for-like quantity, units, configuration and operating context; summarise comparable observations with sample count, range and coverage, applied to verified normal-performance baseline. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not convert temporal association, anomaly scores or symptom similarity into a causal diagnosis or scope extension. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Compare the current symptom against a defensible normal baseline.

MC-037

Category 4

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Has this asset or a genuinely comparable one shown the same symptom before, and which previous actions were followed by verified recovery?

Sensors and telemetry

Proposed evidence: Time-aligned mapped asset telemetry, alarms and traceable portable measurements can compare states only when they represent the same physical quantity and operating context. Coverage: target asset and physically connected comparison points only. Timing: native events plus 1-minute values across the defined before/during/after or peer comparison windows. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify mapping/configuration equivalence, units, calibration/verification, clock, uptime/gaps, drift, placement and response time. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Maintenance-history owner and asset engineer: verified comparable cases, recorded actions and subsequent observed outcomes; similarity criteria must be explicit. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + comparison. Decision interval: during fault triage and after each material new observation or intervention. Minimum evidence: a current controlled source for comparable previous symptom episodes matching the exact job, asset/location and decision time; two like-for-like records/states with common units, known configuration and comparable operating context. Method: resolve exact controlled fields/status and retain source/version; compare like-for-like quantity, units, configuration and operating context, applied to comparable previous symptom episodes. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not convert temporal association, anomaly scores or symptom similarity into a causal diagnosis or scope extension. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Prioritise historically informative checks while re-verifying the present case.

MC-038

Category 4

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Which diagnostic checks have already been completed, under what conditions, with which traceable instruments, and what were the results?

Sensors and telemetry

Proposed evidence: No physical sensor can establish document authority. Telemetry metadata may only corroborate which mapped asset/location produced a reading cited beside a controlled record. Coverage: only the asset/location referenced by the current job document. Timing: source metadata at retrieval time; no physical sampling is required. Quality gate: No sensing is needed to establish the authoritative answer. If contextual telemetry is shown, first verify deployment, asset/location mapping, commissioning, permission, source identity, clock alignment and current health; expose uptime/gaps and provenance. Profile-specific limit: verify point/source ID, mapping version and pipeline health if telemetry is cited. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Work-order owner, contractor quality owner and asset engineer: completed diagnostic steps, instrument IDs/calibration evidence, test conditions and original results. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + comparison. Decision interval: during fault triage and after each material new observation or intervention. Minimum evidence: a current controlled source for diagnostic checks already completed matching the exact job, asset/location and decision time; two like-for-like records/states with common units, known configuration and comparable operating context. Method: resolve exact controlled fields/status and retain source/version; compare like-for-like quantity, units, configuration and operating context, applied to diagnostic checks already completed. Uncertainty: state source/version conflicts, staleness, scope, permission and any ambiguity in asset/location or applicability; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not convert temporal association, anomaly scores or symptom similarity into a causal diagnosis or scope extension. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Avoid repeating tests whose result and conditions are already defensibly recorded.

MC-039

Category 4

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

What are the current priority, response target, authorised work-order scope and explicit exclusions?

Sensors and telemetry

Proposed evidence: No sensing is justified for this decision. If telemetry is displayed as context, it must be limited to the exact asset/work zone and must not be used to infer people, authority, competence, contractual status or permission. Coverage: the exact asset/work zone only, and only when context is genuinely useful. Timing: current state or the shortest decision-relevant interval; no unnecessary historical collection. Quality gate: No sensing is needed to establish the authoritative answer. If contextual telemetry is shown, first verify deployment, asset/location mapping, commissioning, permission, source identity, clock alignment and current health; expose uptime/gaps and provenance. Profile-specific limit: verify mapping, commissioning, permission, clock, health and provenance; no sensor output may substitute for the authoritative record. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Work-order/contract owner and service-desk owner: current priority, response target, authorised scope, exclusions and change-authorisation status. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup. Decision interval: during fault triage and after each material new observation or intervention. Minimum evidence: a current controlled source for current work-order priority/scope/exclusions matching the exact job, asset/location and decision time. Method: resolve exact controlled fields/status and retain source/version, applied to current work-order priority/scope/exclusions. Uncertainty: state source/version conflicts, staleness, scope, permission and any ambiguity in asset/location or applicability; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not convert temporal association, anomaly scores or symptom similarity into a causal diagnosis or scope extension. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Know the authorised scope and exclusions before committing time, parts or intrusive tests.

MC-040

Category 4

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Which next manual test would best distinguish the remaining plausible fault locations while minimising intrusive work?

Sensors and telemetry

Proposed evidence: Proposed mapped plant alarms plus temperature, pressure, flow, current, position or equipment-state telemetry can corroborate current process condition. Coverage: target asset and directly connected upstream/downstream points only. Timing: native alarm events plus 1-minute process data, with 5-minute display summaries where useful. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify point-to-asset mapping, units, calibration/verification, clock offset, uptime, gaps, dead bands, saturation and drift. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Asset engineer and work-order owner: current topology, fault evidence, completed checks and constraints governing permissible next tests. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: diagnosis + recommendation. Decision interval: during fault triage and after each material new observation or intervention. Minimum evidence: a current controlled source for evidence-based fault localisation matching the exact job, asset/location and decision time; a source-labelled fault timeline, current topology and at least one discriminating observation for the retained candidates; only options already permitted by the responsible owner, with explicit ranking criteria. Method: use topology/physics and discriminating evidence to eliminate inconsistent candidates while retaining unresolved alternatives; rank only owner-permitted options by explicit criteria; recommendation creates no permission, applied to evidence-based fault localisation. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not convert temporal association, anomaly scores or symptom similarity into a causal diagnosis or scope extension. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Target the next manual test where it will discriminate most between remaining fault candidates.

MC-041

Category 5

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Which tools, test instruments and access equipment are required for this exact task and installed asset variant?

Sensors and telemetry

Proposed evidence: No physical sensor can establish document authority. Telemetry metadata may only corroborate which mapped asset/location produced a reading cited beside a controlled record. Coverage: only the asset/location referenced by the current job document. Timing: source metadata at retrieval time; no physical sampling is required. Quality gate: No sensing is needed to establish the authoritative answer. If contextual telemetry is shown, first verify deployment, asset/location mapping, commissioning, permission, source identity, clock alignment and current health; expose uptime/gaps and provenance. Profile-specific limit: verify point/source ID, mapping version and pipeline health if telemetry is cited. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Asset engineer, approved task method owner and contractor supervisor: installed variant, specified maintenance procedure and required tool/test/access-equipment capability. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + recommendation. Decision interval: during planning/loading and again before use, delivery, collection or transfer. Minimum evidence: a current controlled source for task-specific tools and access equipment matching the exact job, asset/location and decision time; only options already permitted by the responsible owner, with explicit ranking criteria. Method: resolve exact controlled fields/status and retain source/version; rank only owner-permitted options by explicit criteria; recommendation creates no permission, applied to task-specific tools and access equipment. Uncertainty: state source/version conflicts, staleness, scope, permission and any ambiguity in asset/location or applicability; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not approve unlisted parts, lifting arrangements, unapproved temporary connections or inventory/waste transactions. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Bring the right tools/test gear first time and avoid unsafe improvisation.

MC-042

Category 5

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Which required spares and consumables are confirmed available, where are they held, and are they allocated to this job?

Sensors and telemetry

Proposed evidence: Proposed barcode/RFID scans, smart-shelf weights or container scales can corroborate item presence, quantity or custody after the inventory/waste process has defined the item. Coverage: approved stores, quarantine or transfer locations only. Timing: event scans and current stock/weight snapshot at the transaction time. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify tag-to-item mapping, scale zero/calibration where weight is evidence, clock alignment and transaction gaps. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Stores/inventory owner, asset owner and work-order owner: live stock, reservation/allocation record, approved storage location and stock restrictions. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + observation + comparison. Decision interval: during planning/loading and again before use, delivery, collection or transfer. Minimum evidence: a current controlled source for actual spare/consumable availability matching the exact job, asset/location and decision time; a direct observation or healthy mapped measurement covering the required point/window; two like-for-like records/states with common units, known configuration and comparable operating context. Method: resolve exact controlled fields/status and retain source/version; time-align only relevant observed/measured states and report observation, not cause; compare like-for-like quantity, units, configuration and operating context, applied to actual spare/consumable availability. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not approve unlisted parts, lifting arrangements, unapproved temporary connections or inventory/waste transactions. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Decide what to bring or order before the work window depends on unverified stock.

MC-043

Category 5

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Is the proposed replacement part approved and technically compatible with the installed asset, interfaces, ratings and configuration?

Sensors and telemetry

Proposed evidence: A deliberate QR/NFC/RFID asset-tag read, nameplate check or controller identity can corroborate the exact equipment variant; continuous tracking is not required. Coverage: target asset and its immediate maintainable assembly only. Timing: one-off field read at the workface plus current controller identity where relevant. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify tag-to-asset mapping, identifier legibility, controller/source version and provenance. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Asset engineer, OEM/approved-parts owner, stores/procurement and document controller: installed variant, approved substitutes, ratings and interface constraints. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + comparison. Decision interval: during planning/loading and again before use, delivery, collection or transfer. Minimum evidence: a current controlled source for replacement-part compatibility matching the exact job, asset/location and decision time; two like-for-like records/states with common units, known configuration and comparable operating context. Method: resolve exact controlled fields/status and retain source/version; compare like-for-like quantity, units, configuration and operating context, applied to replacement-part compatibility. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not approve unlisted parts, lifting arrangements, unapproved temporary connections or inventory/waste transactions. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Avoid fitting an incompatible part and know when engineering approval is needed.

MC-044

Category 5

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Which filters, belts, batteries, seals, service kits or other consumables are due or justified by verified condition evidence on this visit?

Sensors and telemetry

Proposed evidence: Proposed mapped plant alarms plus temperature, pressure, flow, current, position or equipment-state telemetry can corroborate current process condition. Coverage: target asset and directly connected upstream/downstream points only. Timing: native alarm events plus 1-minute process data, with 5-minute display summaries where useful. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify point-to-asset mapping, units, calibration/verification, clock offset, uptime, gaps, dead bands, saturation and drift. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Asset engineer, maintenance-strategy owner and stores owner: planned service interval, condition criteria, installed consumable variant and current usage/condition evidence. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: calculation + comparison. Decision interval: during planning/loading and again before use, delivery, collection or transfer. Minimum evidence: a current controlled source for due/condition-justified consumables matching the exact job, asset/location and decision time; two like-for-like records/states with common units, known configuration and comparable operating context; all calculation inputs, units, capacities/tolerances and source versions. Method: show arithmetic, units and tolerance/capacity basis; compare like-for-like quantity, units, configuration and operating context, applied to due/condition-justified consumables. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not approve unlisted parts, lifting arrangements, unapproved temporary connections or inventory/waste transactions. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Bring consumables that are actually due or conditionally justified by the approved strategy.

MC-045

Category 5

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Which delivery point should our vehicle use, and what vehicle, timing, loading and security constraints apply?

Sensors and telemetry

Proposed evidence: Proposed door/lift state, lift load/fault status and temporary laser/ultrasonic clearance measurements can corroborate equipment-route constraints. Coverage: every limiting segment of the already approved route. Timing: state at movement time and one-off clearance measurements at actual constraints. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify route mapping, lift status and clearance-tool accuracy/calibration; state tolerance and temporary-obstruction uncertainty. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Facilities/site-logistics and Security: approved delivery point, vehicle restrictions, booking slot, loading route and security controls. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + recommendation. Decision interval: during planning/loading and again before use, delivery, collection or transfer. Minimum evidence: a current controlled source for delivery bay/vehicle constraints matching the exact job, asset/location and decision time; only options already permitted by the responsible owner, with explicit ranking criteria. Method: resolve exact controlled fields/status and retain source/version; rank only owner-permitted options by explicit criteria; recommendation creates no permission, applied to delivery bay/vehicle constraints. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not approve unlisted parts, lifting arrangements, unapproved temporary connections or inventory/waste transactions. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Send the right vehicle to the right approved delivery point at the right time.

MC-046

Category 5

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Where may we temporarily store tools, spares and removed components, and what environmental, security or fire-loading limits apply?

Sensors and telemetry

Proposed evidence: Environmental sensing is justified only when an approved storage requirement has a defined temperature, humidity or other measurable limit; otherwise use the controlled storage designation alone. Coverage: the approved temporary storage location and the stored item class only. Timing: 1-minute samples shown as 5-minute summaries while condition-sensitive items are stored. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify sensor siting, mapping, units, calibration/verification, clock, uptime, gaps, drift and threshold basis; do not infer security from environmental sensors. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Facilities/site-logistics, Fire Safety and asset/stores owner as applicable: approved temporary storage area, storage conditions, fire-loading and security restrictions. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + observation. Decision interval: during planning/loading and again before use, delivery, collection or transfer. Minimum evidence: a current controlled source for approved temporary storage matching the exact job, asset/location and decision time; a direct observation or healthy mapped measurement covering the required point/window. Method: resolve exact controlled fields/status and retain source/version; time-align only relevant observed/measured states and report observation, not cause, applied to approved temporary storage. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not approve unlisted parts, lifting arrangements, unapproved temporary connections or inventory/waste transactions. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Use an approved temporary store that meets the part/tool constraints.

MC-047

Category 5

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Does this move require lifting equipment, and what approved lift plan, route, capacity and competent roles apply?

Sensors and telemetry

Proposed evidence: Proposed door/lift state, lift load/fault status and temporary laser/ultrasonic clearance measurements can corroborate equipment-route constraints. Coverage: every limiting segment of the already approved route. Timing: state at movement time and one-off clearance measurements at actual constraints. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify route mapping, lift status and clearance-tool accuracy/calibration; state tolerance and temporary-obstruction uncertainty. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Lifting Appointed Person/competent lifting owner, Facilities/logistics and asset owner: approved lift plan, load data, certified equipment and route/capacity constraints. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: calculation + lookup. Decision interval: during planning/loading and again before use, delivery, collection or transfer. Minimum evidence: a current controlled source for lifting need and approved lift arrangements matching the exact job, asset/location and decision time; all calculation inputs, units, capacities/tolerances and source versions. Method: show arithmetic, units and tolerance/capacity basis; resolve exact controlled fields/status and retain source/version, applied to lifting need and approved lift arrangements. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not approve unlisted parts, lifting arrangements, unapproved temporary connections or inventory/waste transactions. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Arrange the right lifting resource and route before equipment arrives.

MC-048

Category 5

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Is each specialist test instrument correctly ranged, functional and within required calibration for the measurement we must make?

Sensors and telemetry

Proposed evidence: Instrument self-test, zero/reference checks and digital calibration metadata can support test-equipment readiness; building telemetry is not evidence of portable-instrument calibration. Coverage: the specific portable instrument and its test setup only. Timing: self-test immediately before use and certificate validity at decision time. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify instrument ID, required range/accuracy, certificate traceability where required, due date, uncertainty and pre-use checks. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Contractor calibration/quality owner and task/asset engineer: instrument identity, calibration evidence, required range/accuracy and pre-use checks. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + comparison. Decision interval: during planning/loading and again before use, delivery, collection or transfer. Minimum evidence: a current controlled source for test-equipment range/function/calibration matching the exact job, asset/location and decision time; two like-for-like records/states with common units, known configuration and comparable operating context. Method: resolve exact controlled fields/status and retain source/version; compare like-for-like quantity, units, configuration and operating context, applied to test-equipment range/function/calibration. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not approve unlisted parts, lifting arrangements, unapproved temporary connections or inventory/waste transactions. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Select defensible test equipment before collecting measurements intended for acceptance.

MC-049

Category 5

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Which temporary power, water, drainage or network connection is approved for contractor use, and what interface limits apply?

Sensors and telemetry

Proposed evidence: Proposed mapped plant alarms plus temperature, pressure, flow, current, position or equipment-state telemetry can corroborate current process condition. Coverage: target asset and directly connected upstream/downstream points only. Timing: native alarm events plus 1-minute process data, with 5-minute display summaries where useful. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify point-to-asset mapping, units, calibration/verification, clock offset, uptime, gaps, dead bands, saturation and drift. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Facilities/service owner, Security/network owner where applicable and work-order owner: approved temporary connection point, capacity/interface limits and connection conditions. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + observation. Decision interval: during planning/loading and again before use, delivery, collection or transfer. Minimum evidence: a current controlled source for approved temporary utility connection matching the exact job, asset/location and decision time; a direct observation or healthy mapped measurement covering the required point/window. Method: resolve exact controlled fields/status and retain source/version; time-align only relevant observed/measured states and report observation, not cause, applied to approved temporary utility connection. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not approve unlisted parts, lifting arrangements, unapproved temporary connections or inventory/waste transactions. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Bring the correct temporary-service arrangements and avoid unapproved connections.

MC-050

Category 5

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

What is the authorised custody route for unused, defective, warranty-return and removed parts after the job?

Sensors and telemetry

Proposed evidence: Proposed barcode/Rfid scans, smart-shelf weights or container scales can corroborate item presence, quantity or custody after the inventory/waste process has defined the item. Coverage: approved stores, quarantine or transfer locations only. Timing: event scans and current stock/weight snapshot at the transaction time. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify tag-to-item mapping, scale zero/calibration where weight is evidence, clock alignment and transaction gaps. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Asset/stores/procurement owner and contract owner: custody, warranty-return, quarantine, reuse and disposal instructions for each part class. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + recommendation. Decision interval: during planning/loading and again before use, delivery, collection or transfer. Minimum evidence: a current controlled source for authorised disposition of removed/unused parts matching the exact job, asset/location and decision time; only options already permitted by the responsible owner, with explicit ranking criteria. Method: resolve exact controlled fields/status and retain source/version; rank only owner-permitted options by explicit criteria; recommendation creates no permission, applied to authorised disposition of removed/unused parts. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not approve unlisted parts, lifting arrangements, unapproved temporary connections or inventory/waste transactions. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Put removed/unused parts into the correct owner-defined custody, return or waste stream.

MC-051

Category 6

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Where are the documented isolation points, and what must the competent isolation process verify before work begins?

Sensors and telemetry

Proposed evidence: Mapped voltage/current, pressure, flow, temperature, valve/position or speed telemetry can corroborate system state but cannot prove dead, zero energy or correct isolation. Coverage: target system, candidate isolation devices and directly connected stored-energy points. Timing: native events and 1-second to 1-minute values through the competent isolation verification sequence. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify mapping, units, calibration/verification, clock, dead bands, shared feeds and loss-of-signal behaviour. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Relevant Authorised Person/service owner and document controller: controlled schematic, isolation schedule, labels, stored-energy information and isolation/proving procedure. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + observation. Decision interval: before the controlled task and immediately after any material condition, permit or isolation change. Minimum evidence: a current controlled source for documented isolation points and required verification matching the exact job, asset/location and decision time; a direct observation or healthy mapped measurement covering the required point/window. Method: resolve exact controlled fields/status and retain source/version; time-align only relevant observed/measured states and report observation, not cause, applied to documented isolation points and required verification. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not declare safe conditions, prove dead/zero energy, select/operate isolations or replace RAMS, permit, competent testing or supervision. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Help the competent isolation process prepare the right points and verification steps without bypassing lockout/proving.

MC-052

Category 6

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Which current hazards and temporary controls apply to this exact workforce, task and time window?

Sensors and telemetry

Proposed evidence: Proposed temperature, humidity, sound level, PM2.5/PM10 and, only where chemically meaningful, VOC sensing can describe local physical conditions without identifying people. Coverage: workforce plus owner-defined adjacent boundary/receptor points only. Timing: 1-minute samples displayed as 5-minute aggregates; preserve exceedance events. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: threshold decisions require task-appropriate calibrated instruments; state range, detection limit, humidity/cross-sensitivity, drift and placement limits. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Health and Safety, local area/service owner and work-order owner: current hazard register/information, temporary controls, active impairments and approved RAMS assumptions. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + observation + comparison. Decision interval: before the controlled task and immediately after any material condition, permit or isolation change. Minimum evidence: a current controlled source for current hazards and temporary controls matching the exact job, asset/location and decision time; a direct observation or healthy mapped measurement covering the required point/window; two like-for-like records/states with common units, known configuration and comparable operating context. Method: resolve exact controlled fields/status and retain source/version; time-align only relevant observed/measured states and report observation, not cause; compare like-for-like quantity, units, configuration and operating context, applied to current hazards and temporary controls. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not declare safe conditions, prove dead/zero energy, select/operate isolations or replace RAMS, permit, competent testing or supervision. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Check whether the approved method still fits current workforce hazards and controls.

MC-053

Category 6

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

What current asbestos information applies to the exact material or location we may disturb?

Sensors and telemetry

Proposed evidence: No IoT sensor, camera or generic air-quality sensor can establish that material contains or does not contain asbestos. Air monitoring belongs only inside a competent asbestos-control process. Coverage: the exact material/disturbance location and any formally specified enclosure monitoring points. Timing: no routine sensing for material identification; accredited sampling/air-monitoring only at the controlled-process cadence. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: rely on accredited sample analysis and calibrated air-monitoring instruments where required; retain sample/location chain of custody. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Asbestos dutyholder/survey owner, Health and Safety and permit/work-order owner; current asbestos register/survey, location-specific investigation and controlled work instructions. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup. Decision interval: before the controlled task and immediately after any material condition, permit or isolation change. Minimum evidence: a current controlled source for location/material-specific asbestos evidence matching the exact job, asset/location and decision time. Method: resolve exact controlled fields/status and retain source/version, applied to location/material-specific asbestos evidence. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not declare material asbestos-free, authorise disturbance or interpret generic air sensing as asbestos detection. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Stop an unsafe asbestos assumption and obtain location-specific competent evidence before disturbance.

MC-054

Category 6

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Does this location require confined-space, restricted-space or other special-entry controls today?

Sensors and telemetry

Proposed evidence: Where the controlled procedure requires it, portable O2, flammable-gas and specified toxic-gas instruments can support atmosphere monitoring; fixed sensors alone do not classify a space. Coverage: exact entry space and the specified pre-entry/continuous sampling positions. Timing: pre-entry value plus continuous or procedure-defined sampling and alarms. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: use maintained, bump-tested/calibrated instruments of the specified type; state response time, sampling method and cross-sensitivity. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Health and Safety/confined-space procedure owner, Facilities/area owner and permit issuer; current space classification, risk assessment, entry permit, rescue plan and restrictions. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + observation. Decision interval: before the controlled task and immediately after any material condition, permit or isolation change. Minimum evidence: a current controlled source for special-entry/confined-space controls matching the exact job, asset/location and decision time; a direct observation or healthy mapped measurement covering the required point/window. Method: resolve exact controlled fields/status and retain source/version; time-align only relevant observed/measured states and report observation, not cause, applied to special-entry/confined-space controls. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not classify/authorise entry to the space or declare its atmosphere safe from sensor readings. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Know whether specialist entry controls, monitoring and rescue arrangements are required.

MC-055

Category 6

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

What electrical system and hazard information must be resolved before we open, test or work near this equipment?

Sensors and telemetry

Proposed evidence: Mapped voltage/current, pressure, flow, temperature, valve/position or speed telemetry can corroborate system state but cannot prove dead, zero energy or correct isolation. Coverage: target system, candidate isolation devices and directly connected stored-energy points. Timing: native events and 1-second to 1-minute values through the competent isolation verification sequence. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify mapping, units, calibration/verification, clock, dead bands, shared feeds and loss-of-signal behaviour. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Electrical Authorised Person/service owner, document controller and Health and Safety: current circuit information, isolation/permit procedure, labels and electrical hazard information. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + observation. Decision interval: before the controlled task and immediately after any material condition, permit or isolation change. Minimum evidence: a current controlled source for electrical hazard and supply information matching the exact job, asset/location and decision time; a direct observation or healthy mapped measurement covering the required point/window. Method: resolve exact controlled fields/status and retain source/version; time-align only relevant observed/measured states and report observation, not cause, applied to electrical hazard and supply information. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not use BMS zeros as proof of dead, authorise live work or infer electrical boundaries without authoritative engineering. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Prepare the correct electrical safety basis before opening/testing equipment.

MC-056

Category 6

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

If the task involves hot work, which permit, alarm-management, fire-watch and post-work controls apply?

Sensors and telemetry

Proposed evidence: No telemetry can show that a permit has been issued. Gas, temperature, pressure, fire-system or electrical measurements are evidence only when the controlled permit/RAMS explicitly requires them. Coverage: the exact workforce and system boundary defined by the permit process. Timing: sampling only at the cadence specified by the controlled process. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: use traceable instruments where specified; building telemetry remains contextual unless the competent permit owner accepts it. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Fire Safety, hot-work permit issuer, Security/control room and local area owner: hot-work controls, fire watch and alarm-management requirements. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + observation. Decision interval: before the controlled task and immediately after any material condition, permit or isolation change. Minimum evidence: a current controlled source for hot-work fire/alarm controls matching the exact job, asset/location and decision time; a direct observation or healthy mapped measurement covering the required point/window. Method: resolve exact controlled fields/status and retain source/version; time-align only relevant observed/measured states and report observation, not cause, applied to hot-work fire/alarm controls. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not issue hot-work permission, disable alarms or replace the fire watch. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Coordinate fire watch, alarm management and permit prerequisites before hot work.

MC-057

Category 6

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Which pressure, temperature, gravity, spring, rotating or other stored-energy hazards must be controlled before opening this system?

Sensors and telemetry

Proposed evidence: Mapped voltage/current, pressure, flow, temperature, valve/position or speed telemetry can corroborate system state but cannot prove dead, zero energy or correct isolation. Coverage: target system, candidate isolation devices and directly connected stored-energy points. Timing: native events and 1-second to 1-minute values through the competent isolation verification sequence. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify mapping, units, calibration/verification, clock, dead bands, shared feeds and loss-of-signal behaviour. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Relevant Authorised Person/service engineer and Health and Safety: current system pressure/temperature state, stored-energy sources, controlled release/verification procedure and limits. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + observation. Decision interval: before the controlled task and immediately after any material condition, permit or isolation change. Minimum evidence: a current controlled source for stored-energy hazards and controls matching the exact job, asset/location and decision time; a direct observation or healthy mapped measurement covering the required point/window. Method: resolve exact controlled fields/status and retain source/version; time-align only relevant observed/measured states and report observation, not cause, applied to stored-energy hazards and controls. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not declare safe conditions, prove dead/zero energy, select/operate isolations or replace RAMS, permit, competent testing or supervision. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Make every relevant stored-energy form explicit before opening the system.

MC-058

Category 6

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Which site-specific controls and current environmental conditions affect our planned work-at-height method?

Sensors and telemetry

Proposed evidence: For roof/external work, local anemometer, rain/wetness, temperature and lighting measurements can support current-condition checks; indoor height work may use measured clearance/level only when relevant. Coverage: the exact roof/external work area or indoor access-equipment setup. Timing: 1-minute weather with 5-minute summaries; task-start and method-defined rechecks. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify sensor siting, calibration/verification, clock and sheltering; authoritative weather warnings and competent inspection take precedence. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Health and Safety, Facilities/roof or area owner and contractor competent person: current work-at-height controls, access-equipment requirements and local weather/surface constraints. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + observation + comparison. Decision interval: before the controlled task and immediately after any material condition, permit or isolation change. Minimum evidence: a current controlled source for work-at-height controls and conditions matching the exact job, asset/location and decision time; a direct observation or healthy mapped measurement covering the required point/window; two like-for-like records/states with common units, known configuration and comparable operating context. Method: resolve exact controlled fields/status and retain source/version; time-align only relevant observed/measured states and report observation, not cause; compare like-for-like quantity, units, configuration and operating context, applied to work-at-height controls and conditions. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not declare safe conditions, prove dead/zero energy, select/operate isolations or replace RAMS, permit, competent testing or supervision. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Decide whether the planned work-at-height method remains suitable for current conditions.

MC-059

Category 6

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Do the conditions we found at the workplace still match the assumptions and controls in the approved RAMS?

Sensors and telemetry

Proposed evidence: No fixed sensor is required. Use deliberate workplace observation and, only where approved, a non-identifying photograph or measured dimension to record current physical condition. Coverage: the target asset/workface and the minimum adjacent area needed to explain the condition. Timing: one-off observation at the decision point; repeat only after a material change. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: record observer role, time, exact location/asset, measuring-tool ID where used and image/file provenance; do not capture people, screens, research content or security-sensitive details. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Contractor RAMS owner/supervisor, Health and Safety and local service/area owner: approved RAMS assumptions plus current field conditions and temporary controls. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: comparison + recommendation. Decision interval: before the controlled task and immediately after any material condition, permit or isolation change. Minimum evidence: a current controlled source for RAMS assumptions versus actual conditions matching the exact job, asset/location and decision time; two like-for-like records/states with common units, known configuration and comparable operating context; only options already permitted by the responsible owner, with explicit ranking criteria. Method: compare like-for-like quantity, units, configuration and operating context; rank only owner-permitted options by explicit criteria; recommendation creates no permission, applied to RAMS assumptions versus actual conditions. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not declare safe conditions, prove dead/zero energy, select/operate isolations or replace RAMS, permit, competent testing or supervision. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Recognise a RAMS-assumption mismatch before work drifts outside the approved method.

MC-060

Category 6

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

If conditions become unsafe, unexpected or outside scope, what is the current stop-work and escalation route?

Sensors and telemetry

Proposed evidence: No sensing is justified for this decision. If telemetry is displayed as context, it must be limited to the exact asset/work zone and must not be used to infer people, authority, competence, contractual status or permission. Coverage: the exact asset/work zone only, and only when context is genuinely useful. Timing: current state or the shortest decision-relevant interval; no unnecessary historical collection. Quality gate: No sensing is needed to establish the authoritative answer. If contextual telemetry is shown, first verify deployment, asset/location mapping, commissioning, permission, source identity, clock alignment and current health; expose uptime/gaps and provenance. Profile-specific limit: verify mapping, commissioning, permission, clock, health and provenance; no sensor output may substitute for the authoritative record. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Contractor supervisor, work-order owner, Health and Safety and duty manager: current stop-work criteria, make-safe expectations, escalation route and scope-change process. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + recommendation. Decision interval: before the controlled task and immediately after any material condition, permit or isolation change. Minimum evidence: a current controlled source for stop-work and escalation route matching the exact job, asset/location and decision time; only options already permitted by the responsible owner, with explicit ranking criteria. Method: resolve exact controlled fields/status and retain source/version; rank only owner-permitted options by explicit criteria; recommendation creates no permission, applied to stop-work and escalation route. Uncertainty: state source/version conflicts, staleness, scope, permission and any ambiguity in asset/location or applicability; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not declare safe conditions, prove dead/zero energy, select/operate isolations or replace RAMS, permit, competent testing or supervision. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Give the crew a clear stop/escalation route when conditions or scope become uncertain.

MC-061

Category 7

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Which tests, tolerances and evidence are required before this task can be presented for technical acceptance?

Sensors and telemetry

Proposed evidence: No physical sensor can establish document authority. Telemetry metadata may only corroborate which mapped asset/location produced a reading cited beside a controlled record. Coverage: only the asset/location referenced by the current job document. Timing: source metadata at retrieval time; no physical sampling is required. Quality gate: No sensing is needed to establish the authoritative answer. If contextual telemetry is shown, first verify deployment, asset/location mapping, commissioning, permission, source identity, clock alignment and current health; expose uptime/gaps and provenance. Profile-specific limit: verify point/source ID, mapping version and pipeline health if telemetry is cited. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Asset/commissioning engineer, work-order owner and OEM/test-plan owner: current acceptance criteria, test sheet, tolerances and witness requirements. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup. Decision interval: before/during the specified test or handover step and after any retest or configuration change. Minimum evidence: a current controlled source for required tests and acceptance evidence matching the exact job, asset/location and decision time. Method: resolve exact controlled fields/status and retain source/version, applied to required tests and acceptance evidence. Uncertainty: state source/version conflicts, staleness, scope, permission and any ambiguity in asset/location or applicability; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not invent acceptance limits, sign certificates, accept residual risk, close permits/work orders or declare handover complete. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Collect exactly the tests/evidence needed so technical completion is not rejected for missing proof.

MC-062

Category 7

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

What pre-work baseline should we preserve so before/after performance can be compared fairly?

Sensors and telemetry

Proposed evidence: Mapped asset/process telemetry plus a traceable spot measurement can preserve comparable pre/post evidence when the same physical quantity, point and operating mode are used. Coverage: target asset and direct context/comparison points only. Timing: preserve raw/source-rate values around the test event and 1- or 5-minute summaries over the agreed stable interval. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify mapping, units, calibration/verification, clock, uptime/gaps, drift and operating-mode comparability. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Asset/commissioning engineer, controls owner and work-order owner: required baseline quantities, test points, operating mode and pre-work evidence-retention rules. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: observation + comparison. Decision interval: before/during the specified test or handover step and after any retest or configuration change. Minimum evidence: a current controlled source for comparable pre-work baseline matching the exact job, asset/location and decision time; a direct observation or healthy mapped measurement covering the required point/window; two like-for-like records/states with common units, known configuration and comparable operating context. Method: time-align only relevant observed/measured states and report observation, not cause; compare like-for-like quantity, units, configuration and operating context, applied to comparable pre-work baseline. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not invent acceptance limits, sign certificates, accept residual risk, close permits/work orders or declare handover complete. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Preserve a comparable before-state so post-work change can be assessed fairly.

MC-063

Category 7

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Which functional or commissioning checks are required after reassembly, restart or return to normal operation?

Sensors and telemetry

Proposed evidence: Proposed mapped plant alarms plus temperature, pressure, flow, current, position or equipment-state telemetry can corroborate current process condition. Coverage: target asset and directly connected upstream/downstream points only. Timing: native alarm events plus 1-minute process data, with 5-minute display summaries where useful. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify point-to-asset mapping, units, calibration/verification, clock offset, uptime, gaps, dead bands, saturation and drift. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Asset/commissioning engineer, OEM/test-plan owner and service operator: current functional-test sequence, operating prerequisites, tolerances and witness requirements. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + observation + comparison. Decision interval: before/during the specified test or handover step and after any retest or configuration change. Minimum evidence: a current controlled source for post-reassembly functional checks matching the exact job, asset/location and decision time; a direct observation or healthy mapped measurement covering the required point/window; two like-for-like records/states with common units, known configuration and comparable operating context. Method: resolve exact controlled fields/status and retain source/version; time-align only relevant observed/measured states and report observation, not cause; compare like-for-like quantity, units, configuration and operating context, applied to post-reassembly functional checks. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not invent acceptance limits, sign certificates, accept residual risk, close permits/work orders or declare handover complete. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Run the required functional checks in the correct sequence and condition.

MC-064

Category 7

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

When can we run the required functional test with low operational disruption and still reproduce the required operating conditions?

Sensors and telemetry

Proposed evidence: Where disruption planning needs demand context, use privacy-preserving area counts or booking/room-status aggregates; do not use identity, device tracking, cameras for recognition or individual trajectories. Coverage: only the affected area and directly dependent public/teaching circulation zones. Timing: 5- or 15-minute aggregates over comparable authorised periods; suppress very small counts where required. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify counter placement/mapping, count error, clock, uptime/gaps and known occlusion/double-counting limits; counts describe area demand, not permission or individual behaviour. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Work-order owner, local area/service scheduler and commissioning engineer: authorised test window, disruption constraints and technical operating conditions required for a valid test. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: forecast + recommendation. Decision interval: before/during the specified test or handover step and after any retest or configuration change. Minimum evidence: a current controlled source for valid low-disruption functional-test timing matching the exact job, asset/location and decision time; at least three comparable prior authorised periods plus current constraints and operating conditions; only options already permitted by the responsible owner, with explicit ranking criteria. Method: use comparable prior periods, exclude known non-comparable periods and return a range; rank only owner-permitted options by explicit criteria; recommendation creates no permission, applied to valid low-disruption functional-test timing. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not invent acceptance limits, sign certificates, accept residual risk, close permits/work orders or declare handover complete. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Select a test time that is both authorised, low disruption and technically valid.

MC-065

Category 7

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Exactly which readings, photographs, certificates and work records must we submit, and which content must not be captured?

Sensors and telemetry

Proposed evidence: Digital instruments may export time-stamped readings; owner-approved photographs can document asset labels/workmanship without people, screens, research content or security-sensitive details. Coverage: maintained asset, permitted test points and permitted work area only. Timing: capture at the task/test moment and preserve original timestamp/file metadata where required. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify instrument/source ID, calibration status, clock offset and file provenance; never fabricate missing metadata. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Work-order/contract owner, acceptance owner and document controller: required readings, photos, certificates, forms, file formats and information-handling restrictions. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + recommendation. Decision interval: before/during the specified test or handover step and after any retest or configuration change. Minimum evidence: a current controlled source for handover evidence package matching the exact job, asset/location and decision time; only options already permitted by the responsible owner, with explicit ranking criteria. Method: resolve exact controlled fields/status and retain source/version; rank only owner-permitted options by explicit criteria; recommendation creates no permission, applied to handover evidence package. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not invent acceptance limits, sign certificates, accept residual risk, close permits/work orders or declare handover complete. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Submit a complete traceable evidence package without capturing restricted information.

MC-066

Category 7

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Are our test readings consistent within expected uncertainty, or does instrument quality, sensor mapping, timing or test condition require a repeat?

Sensors and telemetry

Proposed evidence: Time-aligned mapped asset telemetry, alarms and traceable portable measurements can compare states only when they represent the same physical quantity and operating context. Coverage: target asset and physically connected comparison points only. Timing: native events plus 1-minute values across the defined before/during/after or peer comparison windows. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify mapping/configuration equivalence, units, calibration/verification, clock, uptime/gaps, drift, placement and response time. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Commissioning/asset engineer and contractor quality owner: acceptance tolerance, instrument uncertainty, calibration evidence, mapped building-sensor quality and repeat-test rules. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: comparison + diagnosis. Decision interval: before/during the specified test or handover step and after any retest or configuration change. Minimum evidence: a current controlled source for consistency of test readings matching the exact job, asset/location and decision time; two like-for-like records/states with common units, known configuration and comparable operating context; a source-labelled fault timeline, current topology and at least one discriminating observation for the retained candidates. Method: compare like-for-like quantity, units, configuration and operating context; use topology/physics and discriminating evidence to eliminate inconsistent candidates while retaining unresolved alternatives, applied to consistency of test readings. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not hide discrepant results, auto-adjust sensors or declare an instrument faulty without reference evidence. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Repeat or reconfigure a test when disagreement is explained by measurement quality, timing or placement.

MC-067

Category 7

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Which residual defects, temporary conditions, limitations or follow-up actions must be declared before handover?

Sensors and telemetry

Proposed evidence: Proposed mapped plant alarms plus temperature, pressure, flow, current, position or equipment-state telemetry can corroborate current process condition. Coverage: target asset and directly connected upstream/downstream points only. Timing: native alarm events plus 1-minute process data, with 5-minute display summaries where useful. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify point-to-asset mapping, units, calibration/verification, clock offset, uptime, gaps, dead bands, saturation and drift. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Work-order owner, asset engineer and acceptance owner: current residual defects, temporary conditions, restrictions, owner decisions and follow-up actions. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + observation + comparison. Decision interval: before/during the specified test or handover step and after any retest or configuration change. Minimum evidence: a current controlled source for residual defects and temporary conditions matching the exact job, asset/location and decision time; a direct observation or healthy mapped measurement covering the required point/window; two like-for-like records/states with common units, known configuration and comparable operating context. Method: resolve exact controlled fields/status and retain source/version; time-align only relevant observed/measured states and report observation, not cause; compare like-for-like quantity, units, configuration and operating context, applied to residual defects and temporary conditions. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not invent acceptance limits, sign certificates, accept residual risk, close permits/work orders or declare handover complete. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Declare residual defects/temporary conditions clearly before handover.

MC-068

Category 7

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Which asset, drawing, setting, maintenance or configuration records must be updated because of verified changes we made?

Sensors and telemetry

Proposed evidence: No physical sensor can establish document authority. Telemetry metadata may only corroborate which mapped asset/location produced a reading cited beside a controlled record. Coverage: only the asset/location referenced by the current job document. Timing: source metadata at retrieval time; no physical sampling is required. Quality gate: No sensing is needed to establish the authoritative answer. If contextual telemetry is shown, first verify deployment, asset/location mapping, commissioning, permission, source identity, clock alignment and current health; expose uptime/gaps and provenance. Profile-specific limit: verify point/source ID, mapping version and pipeline health if telemetry is cited. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Asset-data owner, document controller and change-control owner: required asset-register, drawing, settings, maintenance and configuration updates with approval status. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + comparison + recommendation. Decision interval: before/during the specified test or handover step and after any retest or configuration change. Minimum evidence: a current controlled source for controlled-record updates after changes matching the exact job, asset/location and decision time; two like-for-like records/states with common units, known configuration and comparable operating context; only options already permitted by the responsible owner, with explicit ranking criteria. Method: resolve exact controlled fields/status and retain source/version; compare like-for-like quantity, units, configuration and operating context; rank only owner-permitted options by explicit criteria; recommendation creates no permission, applied to controlled-record updates after changes. Uncertainty: state source/version conflicts, staleness, scope, permission and any ambiguity in asset/location or applicability; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not invent acceptance limits, sign certificates, accept residual risk, close permits/work orders or declare handover complete. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Ensure controlled records reflect confirmed changes rather than an outdated installed state.

MC-069

Category 7

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Which authorised role must witness, accept or close each hold point, and what evidence must be available?

Sensors and telemetry

Proposed evidence: No sensing is justified for this decision. If telemetry is displayed as context, it must be limited to the exact asset/work zone and must not be used to infer people, authority, competence, contractual status or permission. Coverage: the exact asset/work zone only, and only when context is genuinely useful. Timing: current state or the shortest decision-relevant interval; no unnecessary historical collection. Quality gate: No sensing is needed to establish the authoritative answer. If contextual telemetry is shown, first verify deployment, asset/location mapping, commissioning, permission, source identity, clock alignment and current health; expose uptime/gaps and provenance. Profile-specific limit: verify mapping, commissioning, permission, clock, health and provenance; no sensor output may substitute for the authoritative record. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Work-order/contract owner, permit issuer and asset/commissioning engineer: current acceptance roles, hold points and evidence prerequisites. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup. Decision interval: before/during the specified test or handover step and after any retest or configuration change. Minimum evidence: a current controlled source for authorised witness/acceptance roles matching the exact job, asset/location and decision time. Method: resolve exact controlled fields/status and retain source/version, applied to authorised witness/acceptance roles. Uncertainty: state source/version conflicts, staleness, scope, permission and any ambiguity in asset/location or applicability; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not invent acceptance limits, sign certificates, accept residual risk, close permits/work orders or declare handover complete. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Sequence witness/acceptance points so the authorised role is available with the right evidence.

MC-070

Category 7

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

What current state, limitations, safeguards and next actions must we communicate to the next operator, shift or maintenance team?

Sensors and telemetry

Proposed evidence: Proposed mapped plant alarms plus temperature, pressure, flow, current, position or equipment-state telemetry can corroborate current process condition. Coverage: target asset and directly connected upstream/downstream points only. Timing: native alarm events plus 1-minute process data, with 5-minute display summaries where useful. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify point-to-asset mapping, units, calibration/verification, clock offset, uptime, gaps, dead bands, saturation and drift. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Service operator, Facilities duty manager and work-order owner: current operating state, limitations, temporary safeguards, outstanding actions and handover recipient/route. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + observation + recommendation. Decision interval: before/during the specified test or handover step and after any retest or configuration change. Minimum evidence: a current controlled source for next-team operational handover matching the exact job, asset/location and decision time; a direct observation or healthy mapped measurement covering the required point/window; only options already permitted by the responsible owner, with explicit ranking criteria. Method: resolve exact controlled fields/status and retain source/version; time-align only relevant observed/measured states and report observation, not cause; rank only owner-permitted options by explicit criteria; recommendation creates no permission, applied to next-team operational handover. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not invent acceptance limits, sign certificates, accept residual risk, close permits/work orders or declare handover complete. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Leave the next operator/shift with current state, limitations and next actions.

MC-071

Category 8

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Which services, spaces and dependent systems could be disrupted by this intervention, and which dependencies are confirmed?

Sensors and telemetry

Proposed evidence: Proposed mapped plant alarms plus temperature, pressure, flow, current, position or equipment-state telemetry can corroborate current process condition. Coverage: target asset and directly connected upstream/downstream points only. Timing: native alarm events plus 1-minute process data, with 5-minute display summaries where useful. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify point-to-asset mapping, units, calibration/verification, clock offset, uptime, gaps, dead bands, saturation and drift. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Service/asset owner, Facilities operations and work-coordination owner: current dependency map, shutdown scope, affected areas and active operational constraints. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + calculation. Decision interval: during coordination/restoration planning and again immediately before the affected activity. Minimum evidence: a current controlled source for dependent services/spaces at risk of disruption matching the exact job, asset/location and decision time; all calculation inputs, units, capacities/tolerances and source versions. Method: resolve exact controlled fields/status and retain source/version; show arithmetic, units and tolerance/capacity basis, applied to dependent services/spaces at risk of disruption. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not authorise shutdown/restoration, remove barriers, reopen areas, classify unknown waste or assign liability/causation. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Plan notification, sequence and contingency for services genuinely dependent on the intervention.

MC-072

Category 8

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Within the approved window, which sub-window is likely to cause the least disruption while still meeting the technical test conditions?

Sensors and telemetry

Proposed evidence: Where disruption planning needs demand context, use privacy-preserving area counts or booking/room-status aggregates; do not use identity, device tracking, cameras for recognition or individual trajectories. Coverage: only the affected area and directly dependent public/teaching circulation zones. Timing: 5- or 15-minute aggregates over comparable authorised periods; suppress very small counts where required. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify counter placement/mapping, count error, clock, uptime/gaps and known occlusion/double-counting limits; counts describe area demand, not permission or individual behaviour. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Work-order owner, local scheduler/area owner and service operator: approved work window, service-demand context, disruption constraints and technical conditions needed for the task. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: forecast + recommendation. Decision interval: during coordination/restoration planning and again immediately before the affected activity. Minimum evidence: a current controlled source for lowest-disruption time inside approved window matching the exact job, asset/location and decision time; at least three comparable prior authorised periods plus current constraints and operating conditions; only options already permitted by the responsible owner, with explicit ranking criteria. Method: use comparable prior periods, exclude known non-comparable periods and return a range; rank only owner-permitted options by explicit criteria; recommendation creates no permission, applied to lowest-disruption time inside approved window. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not authorise shutdown/restoration, remove barriers, reopen areas, classify unknown waste or assign liability/causation. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Choose the least disruptive sub-window inside the approved maintenance window.

MC-073

Category 8

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Which other planned work conflicts with our route, work zone, isolation boundary, shared plant or service?

Sensors and telemetry

Proposed evidence: No sensing is justified for this decision. If telemetry is displayed as context, it must be limited to the exact asset/work zone and must not be used to infer people, authority, competence, contractual status or permission. Coverage: the exact asset/work zone only, and only when context is genuinely useful. Timing: current state or the shortest decision-relevant interval; no unnecessary historical collection. Quality gate: No sensing is needed to establish the authoritative answer. If contextual telemetry is shown, first verify deployment, asset/location mapping, commissioning, permission, source identity, clock alignment and current health; expose uptime/gaps and provenance. Profile-specific limit: verify mapping, commissioning, permission, clock, health and provenance; no sensor output may substitute for the authoritative record. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Work-coordination owner, permit issuer and Facilities operations: current planned work, route bookings, exclusion zones, isolation boundaries and shared-plant conflicts. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + comparison. Decision interval: during coordination/restoration planning and again immediately before the affected activity. Minimum evidence: a current controlled source for other work conflicting by area/service/isolation matching the exact job, asset/location and decision time; two like-for-like records/states with common units, known configuration and comparable operating context. Method: resolve exact controlled fields/status and retain source/version; compare like-for-like quantity, units, configuration and operating context, applied to other work conflicting by area/service/isolation. Uncertainty: state source/version conflicts, staleness, scope, permission and any ambiguity in asset/location or applicability; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not authorise shutdown/restoration, remove barriers, reopen areas, classify unknown waste or assign liability/causation. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Avoid two teams competing for the same route, isolation, plant or exclusion zone.

MC-074

Category 8

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

What containment and monitoring are required for noise, dust, debris, vibration or odour from this job?

Sensors and telemetry

Proposed evidence: Proposed temperature, humidity, sound level, PM2.5/PM10 and, only where chemically meaningful, VOC sensing can describe local physical conditions without identifying people. Coverage: workplace plus owner-defined adjacent boundary/receptor points only. Timing: 1-minute samples displayed as 5-minute aggregates; preserve exceedance events. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: threshold decisions require task-appropriate calibrated instruments; state range, detection limit, humidity/cross-sensitivity, drift and placement limits. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Health and Safety, local area owner and work-order owner: current containment requirements, nuisance limits where defined, monitoring method and escalation thresholds. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + observation + comparison. Decision interval: during coordination/restoration planning and again immediately before the affected activity. Minimum evidence: a current controlled source for noise/dust/debris/odour controls matching the exact job, asset/location and decision time; a direct observation or healthy mapped measurement covering the required point/window; two like-for-like records/states with common units, known configuration and comparable operating context. Method: resolve exact controlled fields/status and retain source/version; time-align only relevant observed/measured states and report observation, not cause; compare like-for-like quantity, units, configuration and operating context, applied to noise/dust/debris/odour controls. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not authorise shutdown/restoration, remove barriers, reopen areas, classify unknown waste or assign liability/causation. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Use appropriate containment/monitoring and respond to measured effects without overclaiming exposure.

MC-075

Category 8

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Which approved waste stream should each material enter, and what segregation, storage, transfer and evidence are required?

Sensors and telemetry

Proposed evidence: No sensor should classify waste chemistry. Barcode/RFID identity, container state and calibrated scales may support quantity/custody after classification by the responsible process. Coverage: approved waste containers, quarantine point and transfer area only. Timing: event scans and one-off weights at transfer. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify item/container mapping, scale calibration when weight is evidential, clock and transaction completeness. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Waste/environmental manager, contract owner, Health and Safety/COSHH owner and stores: material classification, approved waste route, segregation and transfer records. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + calculation. Decision interval: during coordination/restoration planning and again immediately before the affected activity. Minimum evidence: a current controlled source for waste stream and transfer evidence matching the exact job, asset/location and decision time; all calculation inputs, units, capacities/tolerances and source versions. Method: resolve exact controlled fields/status and retain source/version; show arithmetic, units and tolerance/capacity basis, applied to waste stream and transfer evidence. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not authorise shutdown/restoration, remove barriers, reopen areas, classify unknown waste or assign liability/causation. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Segregate and document waste correctly and quarantine unknown material rather than guessing.

MC-076

Category 8

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Which barriers, signs, exclusion zones or temporary route diversions are required, while preserving egress and accessibility?

Sensors and telemetry

Proposed evidence: No sensing is justified for this decision. If telemetry is displayed as context, it must be limited to the exact asset/work zone and must not be used to infer people, authority, competence, contractual status or permission. Coverage: the exact asset/work zone only, and only when context is genuinely useful. Timing: current state or the shortest decision-relevant interval; no unnecessary historical collection. Quality gate: No sensing is needed to establish the authoritative answer. If contextual telemetry is shown, first verify deployment, asset/location mapping, commissioning, permission, source identity, clock alignment and current health; expose uptime/gaps and provenance. Profile-specific limit: verify mapping, commissioning, permission, clock, health and provenance; no sensor output may substitute for the authoritative record. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Facilities/site-logistics, Security, Fire Safety and accessibility/area owner as applicable: approved barriers, signage, route diversions, egress and accessibility constraints. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + recommendation. Decision interval: during coordination/restoration planning and again immediately before the affected activity. Minimum evidence: a current controlled source for barriers/signage/diversion requirements matching the exact job, asset/location and decision time; only options already permitted by the responsible owner, with explicit ranking criteria. Method: resolve exact controlled fields/status and retain source/version; rank only owner-permitted options by explicit criteria; recommendation creates no permission, applied to barriers/signage/diversion requirements. Uncertainty: state source/version conflicts, staleness, scope, permission and any ambiguity in asset/location or applicability; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not authorise shutdown/restoration, remove barriers, reopen areas, classify unknown waste or assign liability/causation. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Protect others while keeping required access, accessibility and egress workable.

MC-077

Category 8

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

What is the authorised service-restoration sequence, and which prerequisites and dependencies must be confirmed before each stage?

Sensors and telemetry

Proposed evidence: Mapped voltage/current, pressure, flow, temperature, valve/position or speed telemetry can corroborate system state but cannot prove dead, zero energy or correct isolation. Coverage: target system, candidate isolation devices and directly connected stored-energy points. Timing: native events and 1-second to 1-minute values through the competent isolation verification sequence. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify mapping, units, calibration/verification, clock, dead bands, shared feeds and loss-of-signal behaviour. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Relevant Authorised Person/service operator, asset engineer and permit/work-order owner: controlled restoration sequence, dependencies, hold points and permit conditions. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: lookup + observation + comparison. Decision interval: during coordination/restoration planning and again immediately before the affected activity. Minimum evidence: a current controlled source for authorised restoration sequence and prerequisites matching the exact job, asset/location and decision time; a direct observation or healthy mapped measurement covering the required point/window; two like-for-like records/states with common units, known configuration and comparable operating context. Method: resolve exact controlled fields/status and retain source/version; time-align only relevant observed/measured states and report observation, not cause; compare like-for-like quantity, units, configuration and operating context, applied to authorised restoration sequence and prerequisites. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not authorise shutdown/restoration, remove barriers, reopen areas, classify unknown waste or assign liability/causation. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Support a deliberate owner-controlled restoration sequence with visible prerequisites.

MC-078

Category 8

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

After restoration, what operating conditions and observation period are required before the system can be presented for handover?

Sensors and telemetry

Proposed evidence: Mapped asset/process telemetry plus a traceable spot measurement can preserve comparable pre/post evidence when the same physical quantity, point and operating mode are used. Coverage: target asset and direct context/comparison points only. Timing: preserve raw/source-rate values around the test event and 1- or 5-minute summaries over the agreed stable interval. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify mapping, units, calibration/verification, clock, uptime/gaps, drift and operating-mode comparability. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Asset/commissioning engineer, service operator and work-order owner: required post-restoration operating mode, stability criteria, observation duration and handover hold point. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: observation + comparison. Decision interval: during coordination/restoration planning and again immediately before the affected activity. Minimum evidence: a current controlled source for post-restoration stability observation matching the exact job, asset/location and decision time; a direct observation or healthy mapped measurement covering the required point/window; two like-for-like records/states with common units, known configuration and comparable operating context. Method: time-align only relevant observed/measured states and report observation, not cause; compare like-for-like quantity, units, configuration and operating context, applied to post-restoration stability observation. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not authorise shutdown/restoration, remove barriers, reopen areas, classify unknown waste or assign liability/causation. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Know whether enough valid stability evidence exists to proceed to competent handover.

MC-079

Category 8

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

Has the work area been returned to the clean, accessible and operational condition specified for owner handback?

Sensors and telemetry

Proposed evidence: No fixed sensor is required. Use deliberate workplace observation and, only where approved, a non-identifying photograph or measured dimension to record current physical condition. Coverage: the target asset/workface and the minimum adjacent area needed to explain the condition. Timing: one-off observation at the decision point; repeat only after a material change. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: record observer role, time, exact location/asset, measuring-tool ID where used and image/file provenance; do not capture people, screens, research content or security-sensitive details. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Facilities/area owner and work-order owner; specified handback condition for cleanliness, access, barriers, reinstatement, leaks/damage and operational readiness. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: observation + comparison. Decision interval: during coordination/restoration planning and again immediately before the affected activity. Minimum evidence: a current controlled source for work-area handback condition matching the exact job, asset/location and decision time; a direct observation or healthy mapped measurement covering the required point/window; two like-for-like records/states with common units, known configuration and comparable operating context. Method: time-align only relevant observed/measured states and report observation, not cause; compare like-for-like quantity, units, configuration and operating context, applied to work-area handback condition. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not authorise shutdown/restoration, remove barriers, reopen areas, classify unknown waste or assign liability/causation. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Find incomplete housekeeping/access issues before the team leaves.

MC-080

Category 8

EXTERNAL MAINTENANCE CONTRACTORS QUESTION

If a new issue is reported after we leave, what evidence can distinguish a plausible relationship to our intervention from coincidence without assigning blame?

Sensors and telemetry

Proposed evidence: Time-aligned mapped asset telemetry, alarms and traceable portable measurements can compare states only when they represent the same physical quantity and operating context. Coverage: target asset and physically connected comparison points only. Timing: native events plus 1-minute values across the defined before/during/after or peer comparison windows. Quality gate: Treat sensing/integrations as proposed until deployment, asset/location mapping, commissioning and permission are independently verified. For evidence use, record modality/point ID, units, native sampling and displayed aggregation; verify calibration or functional check as applicable, common-clock alignment, uptime/gaps, drift/range flags and provenance. Profile-specific limit: verify mapping/configuration equivalence, units, calibration/verification, clock, uptime/gaps, drift, placement and response time. If the quality gate fails, treat the measurement as unavailable or clearly contextual, not as a hidden substitute. Privacy: use asset/area evidence only; do not identify or track people.

Authoritative supporting data

Owner/source: Work-order/contract owner, asset engineer and service operator; original pre/post evidence, intervention record, subsequent fault timeline and any independent diagnostic findings. Precedence: live emergency, safety, permit or Authorised Person decisions > current controlled owner records > verified direct observations/commissioned telemetry > uncontrolled copies > historical reference. Time/permission: show effective time, observed time for physical state, retrieved time, owner, version/status, requester permission and provenance. Staleness/conflict: treat evidence as stale when superseded, expired, outside its stated scope, or when its effective/observed time no longer covers the decision interval. Keep conflicts visible and apply the higher authority; unresolved equal-authority conflict returns "not established". The May 2021 floor plan is historical spatial orientation only and cannot establish current room use, access, services, hazards or closures.

Analysis required

Operation: comparison + diagnosis. Decision interval: during coordination/restoration planning and again immediately before the affected activity. Minimum evidence: a current controlled source for post-work issue relationship to intervention matching the exact job, asset/location and decision time; two like-for-like records/states with common units, known configuration and comparable operating context; a source-labelled fault timeline, current topology and at least one discriminating observation for the retained candidates. Method: compare like-for-like quantity, units, configuration and operating context; use topology/physics and discriminating evidence to eliminate inconsistent candidates while retaining unresolved alternatives, applied to post-work issue relationship to intervention. Uncertainty: state source/version conflicts, staleness, scope and permission; plus asset/location mapping, calibration/verification, clock offset, uptime/gaps, drift/range or detection limits, placement, sample coverage and relevant confounding; do not hide disagreement inside a single confidence score. Safe failure: if minimum evidence is missing, stale, conflicting, outside permission or technically unhealthy, return "not established", name the blocking gap and responsible current owner/competent role, and do not fill it with historical assumptions or unsupported inference.

Answer boundary

Do not assign blame, accept/reject liability or state causal responsibility without competent engineering evidence. Information is not permission. The service must not authorise work, access, permits or isolations; operate access, BMS or plant controls; replace risk assessment, RAMS, lockout/proving or competent supervision; infer competence or personal behaviour; or disclose personal, security-sensitive or commercially restricted information.

Why this is useful

Support fair technical triage of a post-work issue without premature blame or causal attribution.

Summary: what a defensible contractor answer should do

The catalogue is useful only when evidence quality and authority travel with the answer.

Ask the operational question

Each MC record ends in a real decision: dispatch, route, verify an asset, select a diagnostic check, bring kit, stop/escalate, test, coordinate or hand over.

Prefer authority over inference

Current competent-person, permit, safety, engineering and operational decisions outrank telemetry. Controlled owner records outrank uncontrolled copies and historical material.

Verify the evidence chain

For proposed sensing, verify deployment, mapping, commissioning, permission, modality/location, sampling/aggregation, calibration/verification, clock alignment, uptime, gaps, drift and provenance before use.

Use the right operation

Do not call every output analysis. Distinguish lookup, observation, calculation, estimate, forecast, comparison, diagnosis and recommendation. Each has a different minimum-evidence burden.

Keep uncertainty visible

A healthy measurement can still be incomplete; a trend can be associated but not causal; a forecast can rank options but not promise conditions; a historical plan can orient but not authorise.

Fail safely

If minimum evidence is missing, stale, conflicting, technically unhealthy or outside permission, answer "not established", explain the blocking gap and send the decision to the responsible current owner.

Preserve safety processes

Nothing in the service issues access, permits or isolations; proves dead/zero energy; replaces RAMS/risk assessment; certifies competence/compliance; or operates building controls.

Protect people and sensitive information

Use asset/area-level evidence. Do not identify/track people, inspect speech/research/code/network content, infer productivity/competence from behaviour, or expose personal/security/commercially restricted information.